Unit-3

IOT Data link layer and Network layer protocols

- **PHY/MAC Layer**
  - 3GPP MTC
  - IEEE 802.11
  - IEEE 802.15
  - Wireless HART
  - Z Wave
  - Bluetooth Low Energy
  - Zigbee Smart Energy
  - DASH7

- **Network Layer**
  - IPv4
  - IPv6
  - 6LoWPAN
  - 6TiSCH
  - ND
  - DHCP
  - ICMP
  - RPL
  - CORPL
  - CARP

IOT Data link layer and Network layer protocols

PHY Layer:

- physical transmission of data
- hardware
- electrical specifications
- modulation,
- coding,
- frequency bands,
- transmission power, and
- signal propagation

IOT Data link layer and Network layer protocols

MAC Layer:

- timing
- synchronization
- channel access methods
- avoiding collisions
- efficient utilization of the available bandwidth

IOT Data link layer and Network layer protocols

MAC Layer:

- Channel accessing Methods
  - contention based access (e.g., CSMA/CA - Carrier Sense Multiple Access with Collision Avoidance),
  - time-based access (e.g., TDMA - Time Division Multiple Access), or
  - scheduled access (e.g., FDMA - Frequency Division Multiple Access)

IOT Data link layer and Network layer protocols

MAC Layer:

the MAC layer may handle functions such as

- packet fragmentation,
- error detection,
- acknowledgments, and
- retransmissions

IOT Data link layer and Network layer protocols

1. 3GPP MTC

What is 3GPP ?

ARIB
www.arib.or.jp
The Association of Radio Industries and Businesses, Japan

ATIS
www.atis.org
The Alliance for Telecommunications Industry Solutions, USA

CCSA
www.ccsa.org.cn
China Communications Standards Association

ETSI
www.etsi.org
The European Telecommunications Standards Institute

TSDSI
http://tsdsi.org/
Telecommunications Standards Development Society, India



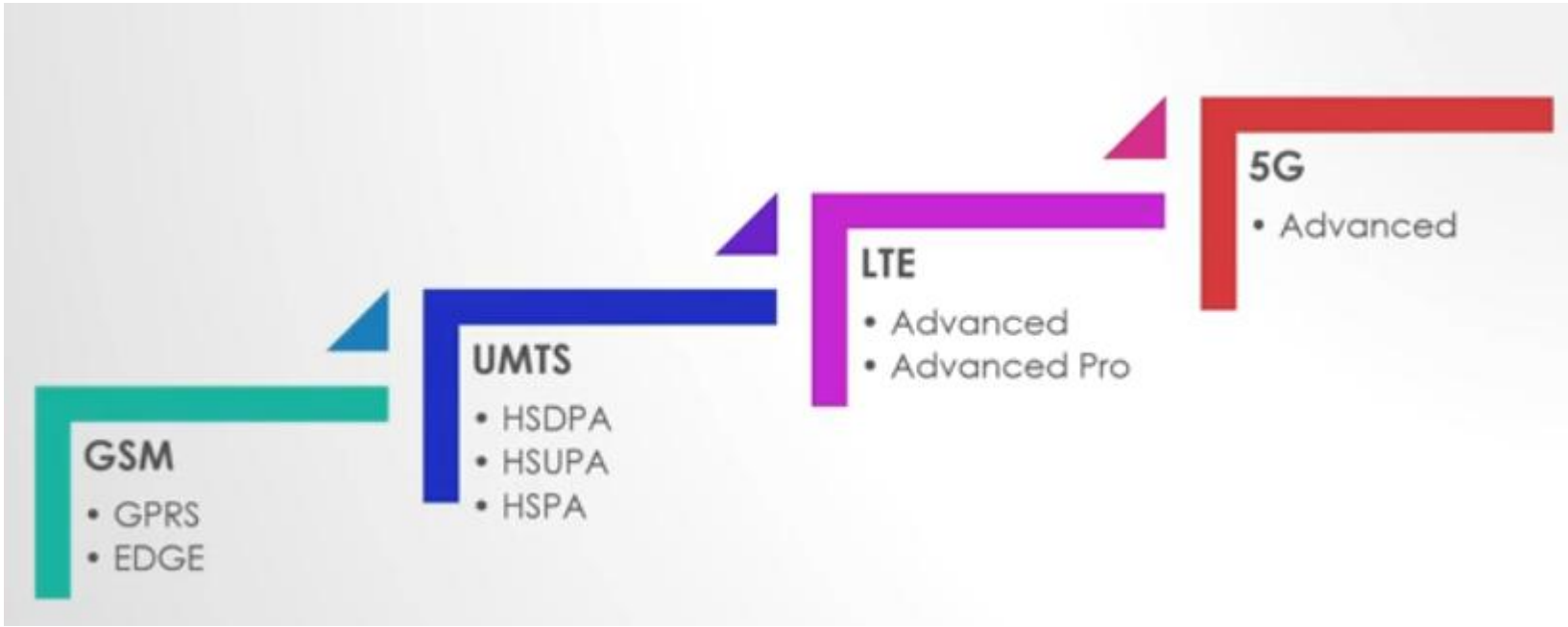
TTA
www.tta.or.kr
Telecommunications Technology Association, Korea

TTC
www.ttc.or.jp
Telecommunication Technology Committee, Japan

IOT Data link layer and Network layer protocols

1. 3GPP MTC

3GPP Supports



IOT Data link layer and Network layer protocols

1. 3GPP MTC



- 3GPP (3rd Generation Partnership Project) MTC (Machine-Type Communications) is designed to support communication between machines, like IoT devices, over cellular networks.

IOT Data link layer and Network layer protocols

1. 3GPP MTC

Key aspects of 3GPP MTC in IoT

i. **Low-Power Consumption**

ii. **Enhanced Coverage**

iii. **Support for Massive Device Connectivity**

iv. **Low-Cost Device Design**

v. **Reduced Data Transmission Requirements**

vi. **Enhanced Security for IoT**

vii. **Efficient Network Resource Usage**

viii. **Improved Latency Control**

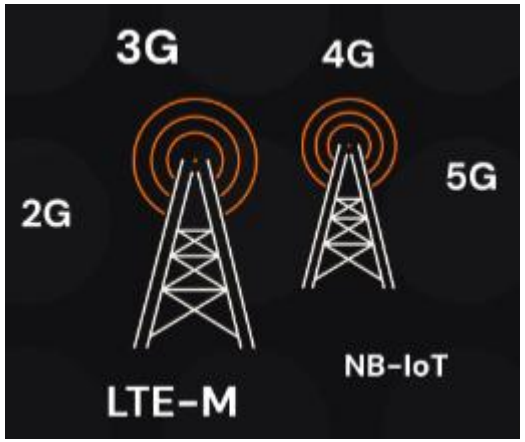
ix. **Support for Mobility and Stationary Devices**

IOT Data link layer and Network layer protocols

1. 3GPP MTC

Key 3GPP Technologies Under MTC:

- **LTE-M**: (Long Term Evoluation for Machines) :Designed to provide low-power, wide-area connectivity within LTE networks.
- **NB-IoT**: A narrowband technology that supports extended coverage and low data rate, making it ideal for low-cost and low-power IoT devices.



IOT Data link layer and Network layer protocols

1. 3GPP MTC



What are the key features of LTE-M?

FEATURE	LTE-M PROTOCOL
Deployment	In-band LTE
Minimum carrier bandwidth	1400 kHz
Data rate	Up to 1000 kbps
Transmission range	5 km
Security	AES 256
Modulation	QPSK, 16-QAM and 64-QAM
Duplexing	FD & HD (type B), FDD & TDD
Power saving	PSM, ext. I-DRX, C-DRX
Power class	23 dBm, 20 dBm
Battery life time	10 years
Link budget	146 db

IOT Data link layer and Network layer protocols

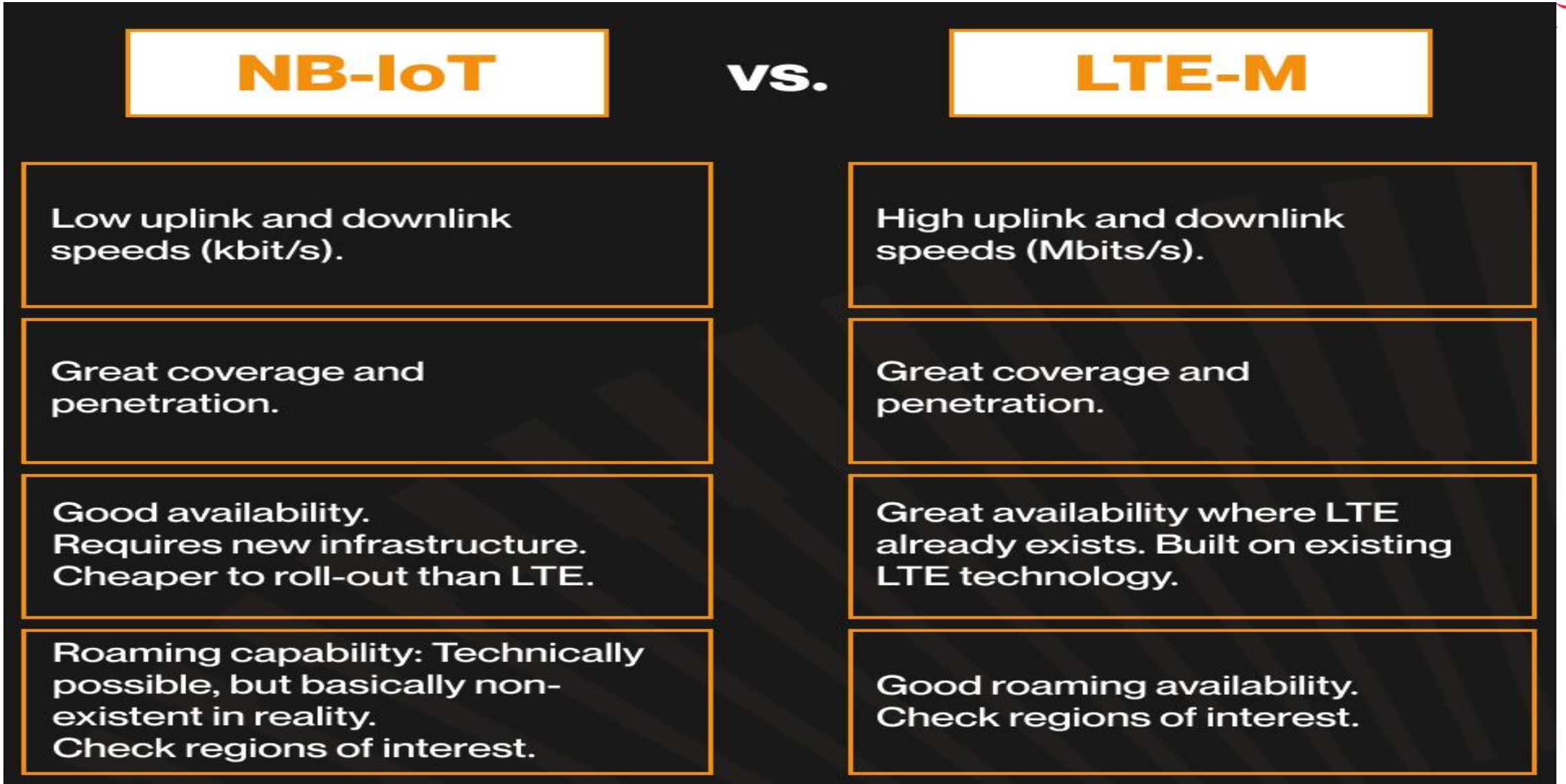
1. 3GPP MTC

NB-IoT evolution: LTE Cat-NB1 vs Cat-NB2

Feature	LTE Cat-NB1 (NB-IoT release 13)	LTE Cat-NB2 (NB-IoT release 14)
Uplink data rate	Up to 66 kbps	Up to 159 kbps
Downlink data rate	Up to 26 kbps	Up to 127 kbps
Bandwidth	180 KHz carrier	180 kHz carrier with improved modulation
Power consumption	Very low, supports PSM and eDRX	Low, supports PSM and eDRX with enhancements
Coverage	Good, deep indoor and rural areas	Good, with improved performance in challenging environments
Latency	Higher	Lower
Efficiency	Standard	Enhanced
Positioning	Basic (Cell-ID)	Enhanced (OTDOA, E-CID)

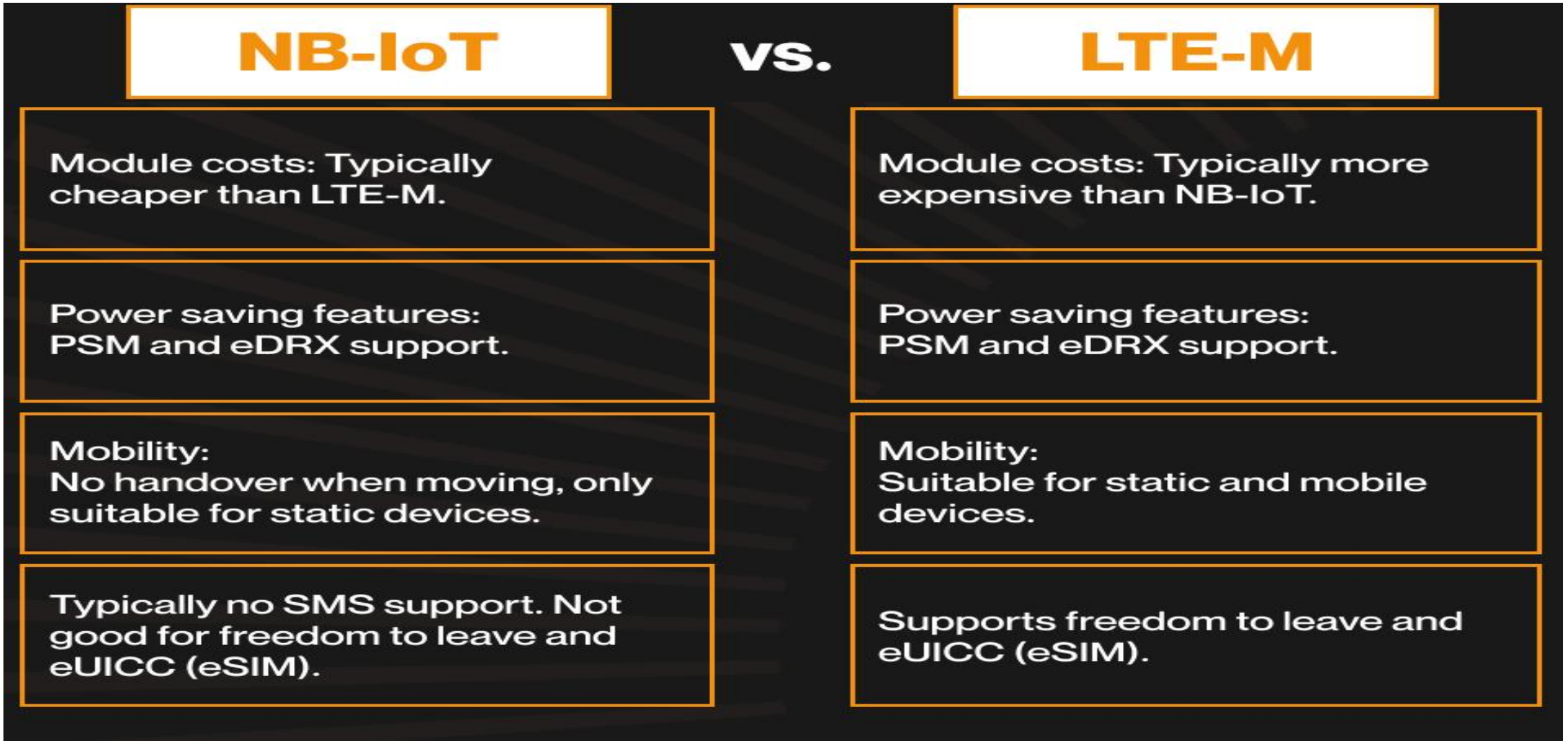
IOT Data link layer and Network layer protocols

1. 3GPP MTC



IOT Data link layer and Network layer protocols

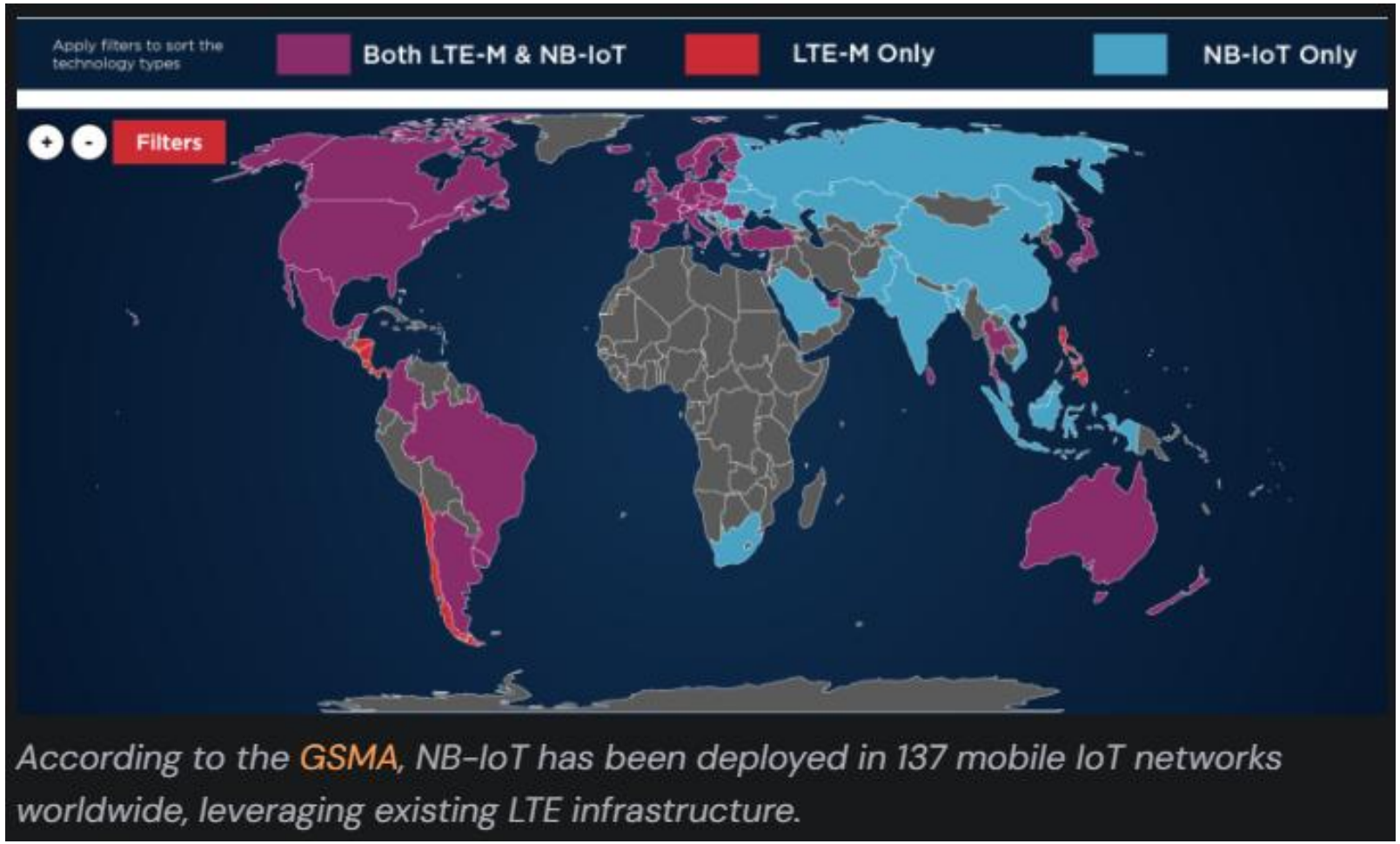
1. 3GPP MTC



IOT Data link layer and Network layer protocols

1. 3GPP MTC

Global distribution map of mobile connectivity

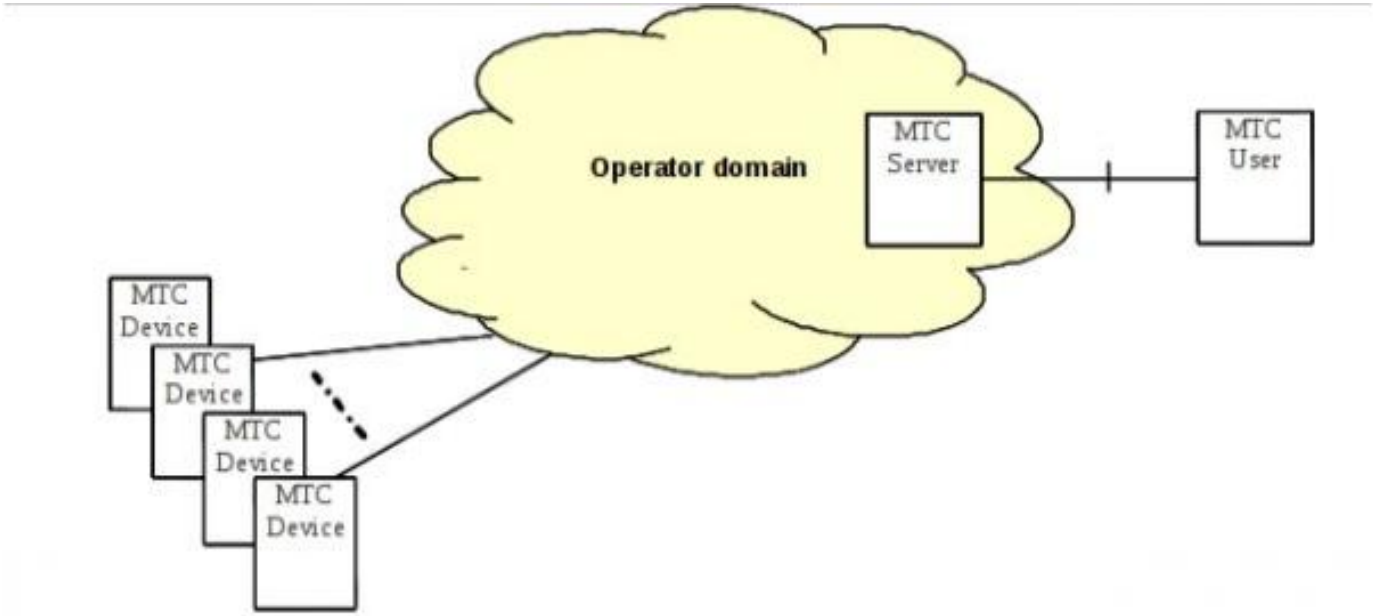


IOT Data link layer and Network layer protocols

1. 3GPP MTC

MTC communication scenarios

1) Communication scenario with MTC devices communicating with MTC server. MTC server is located in the operator domain



3GPP MTC Architecture-MTC server located in the operated domain

IOT Data link layer and Network layer protocols

1. 3GPP MTC

MTC communication scenarios

2) Communication scenario with MTC devices communicating with MTC server. MTC server is located outside the operator domain:

The network operator provides network connectivity to MTC Server(s). This applies to MTC Servers controlled by the network operator or to MTC Servers not controlled by the network operator



3GPP MTC Architecture-MTC server located outside the operated domain

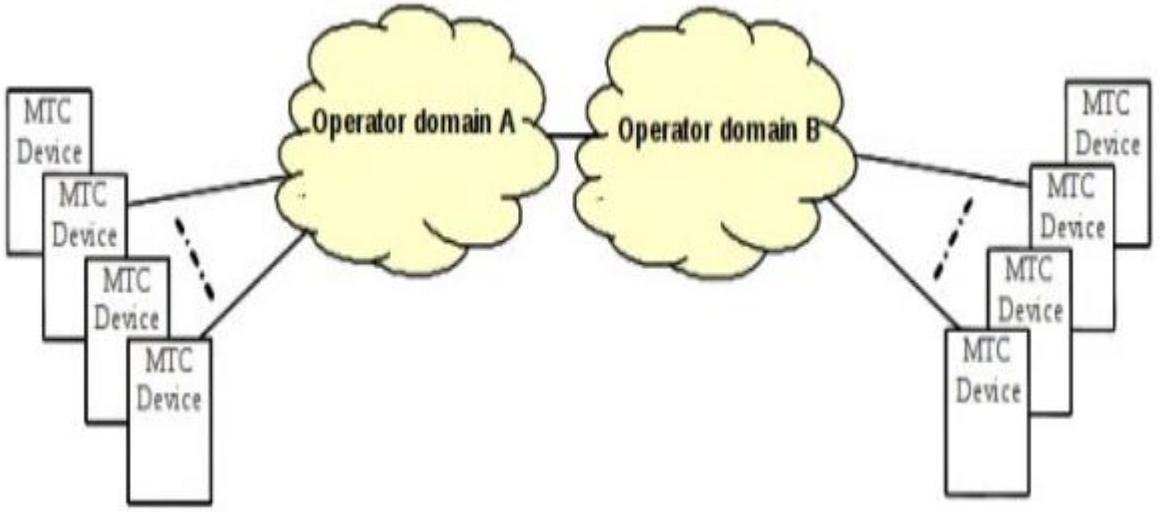
IOT Data link layer and Network layer protocols

1. 3GPP MTC

MTC communication scenarios

3) MTC devices communicating with each other: The communication scenario where the

MTC Devices communicate directly without intermediate MTC Server



IOT Data link layer and Network layer protocols

1. 3GPP MTC

MTC communication scenarios

- MTC Device: A MTC Device is a UE equipped for Machine Type Communication, which communicates through a PLMN with MTC Server(s) and/or other MTC Device(s).
- MTC Feature: MTC Features are network functions to optimise the network for use by M2M applications.
- MTC Group: A MTC Group is a group of MTC Devices that share one or more MTC Features and that belong to the same MTC Subscriber.

IOT Data link layer and Network layer protocols

1. 3GPP MTC

MTC communication scenarios

- MTC Server: A MTC Server is a server, which communicates to the PLMN itself, and to MTC Devices through the PLMN. The MTC Server can also have an interface which can be accessed by the MTC User. The MTC Server can:
- Provides services for other servers (e.g. The MTC Server is a Services Capability Server for an Application Server), and/or Provides services for applications and can host the application (e.g. The MTC Server is an Application Server)

IOT Data link layer and Network layer protocols

1. 3GPP MTC

MTC communication scenarios

- MTC User: A MTC User uses the service provided by the MTC Server.
- MTC Subscriber: A MTC Subscriber is a legal entity having a contractual relationship with the network operator to provide service to one or more MTC Devices



IOT Data link layer and Network layer protocols

1. 3GPP MTC

PHY layer features

- **Narrowband Operation:** LTE-M operates within a restricted 1.4 MHz channel , whereas NB-IoT scales down to 180 kHz.
- **Coverage Enhancement (CE) via Subframe Repetition:** Delivers a 15–20 dB link budget gain over standard LTE to reach deep indoors and underground.
- **Half-Duplex FDD (HD-FDD):** By disabling RX while TX is active, devices remove the need for costly and space-consuming RF duplexers.
- **Lower-Order Modulation:** Uses robust, simplified modulation schemes like BPSK and QPSK.
- **Reduced Transmit Power:** Supports lower maximum output power classes (such as 14 dBm or 20 dBm instead of standard 23 dBm) to avoid peak current spikes on small batteries.

IOT Data link layer and Network layer protocols

1. 3GPP MTC

MAC layer features

- **Power Saving Mode (PSM):** Allows a device to turn off its radio transceiver and enter a deep sleep state for days or weeks.
- **Extended Discontinuous Reception (eDRX):** Lengthens the paging listening cycles up to 40 minutes for LTE-M (and several hours for NB-IoT), letting the device sleep soundly between scheduled downlinks while staying reachable.
- **Early Data Transmission (EDT):** Permits small data payloads to be included directly inside the 4-step Random Access Channel (RACH) handshakes.
- **Repetition Scheduling & CE Modes:** The MAC layer dynamically manages frame scheduling based on Coverage Enhancement levels.

IOT Data link layer and Network layer protocols

1. 3GPP MTC

Hardware Modules

- **Quectel**: BG95-M3, BG96
- **SIMCom**: SIM7080G, SIM7000G
- **Nordic Semiconductor**: nRF9160, nRF9161
- **u-blox**: SARA-R5, SARA-R4
- **Telit Cinterion**: ME910G1, EXS82
- **Sierra Wireless**: AirPrime HL7800



Quectel BG95-M3

IOT Data link layer and Network layer protocols

1. 3GPP MTC



SIM7080G



SIM7000E

IOT Data link layer and Network layer protocols

IEEE 802.11

What is IEEE 802.11 ?





IOT Data link layer and Network layer protocols

IEEE 802.11

Why is IEEE 802.11 ?

- This standard is a key IoT wireless access technology,
- either for connecting endpoints such as fog computing nodes,
- high-data-rate sensors, and
- audio or video analytics devices or
- for deploying Wi-Fi backhaul infrastructures, such as
- outdoor Wi-Fi mesh in smart cities, oil and mining, or other environments.
- However, Wi-Fi lacks sub-GHz support for better signal penetration, low power for battery-powered nodes, and the ability to support a large number of devices.

IOT Data link layer and Network layer protocols

IEEE 802.11

- IEEE 802.11 standard, popularly known as WiFi, lays down the architecture and specifications of wireless LANs (WLANs).
- WiFi or WLAN uses high frequency radio waves for connecting the nodes.

- IEEE 802.11 is a set of standards for wireless LAN technology created by the Institute of Electrical and Electronics Engineers (IEEE)

- The original 802.11 standard was released in 1997



IOT Data link layer and Network layer protocols

IEEE 802.11

There are several standards of IEEE 802.11 WLANs.

The prominent among them are

802.11,

802.11a,

802.11b,

802.11g,

802.11n and

802.11p.



- All the standards use carrier-sense multiple access with collision avoidance (CSMA/CA).
- Also, they have support for both centralized base station based as well as ad-hoc networks

IOT Data link layer and Network layer protocols

IEEE 802.11

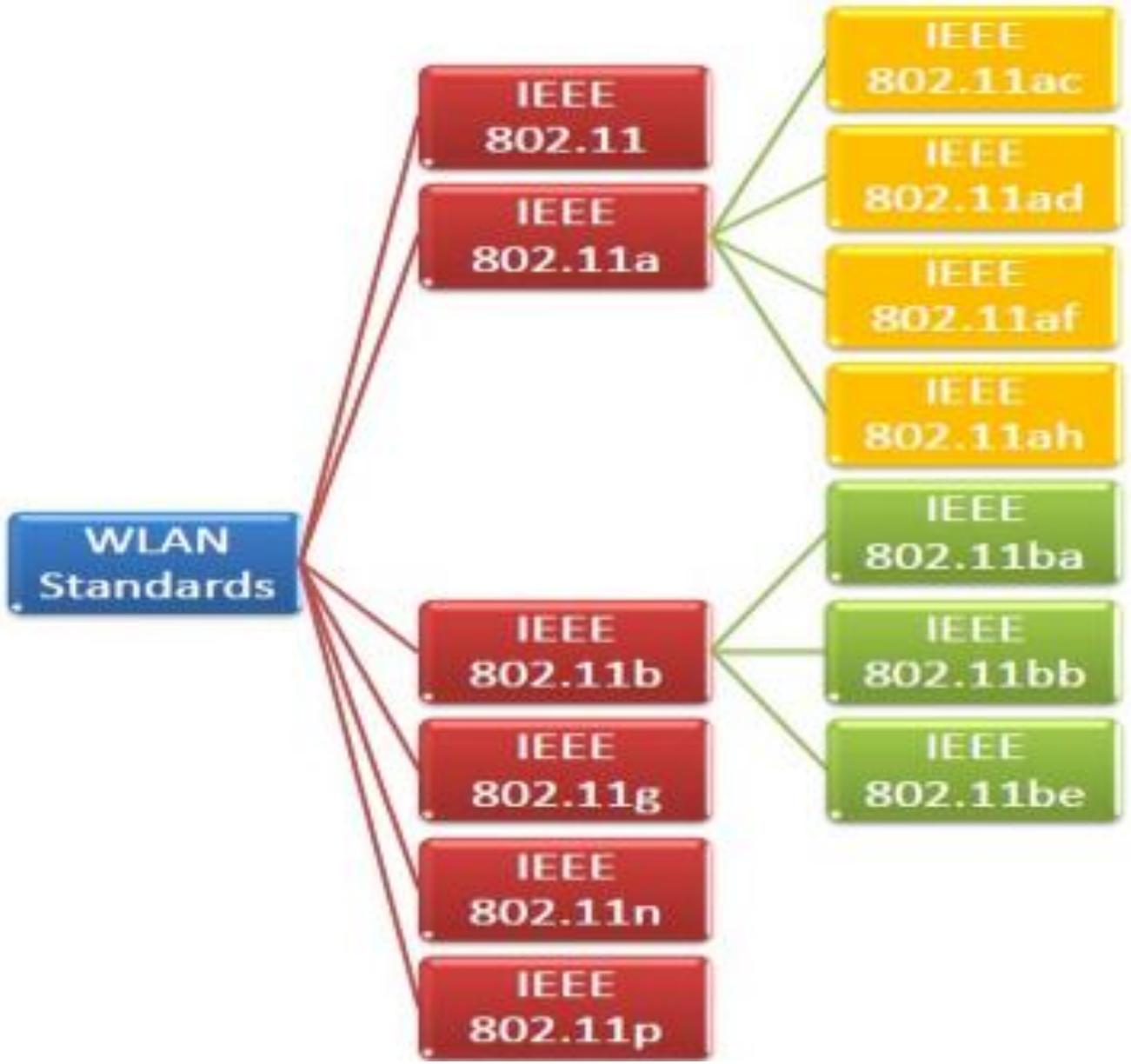


Evolution of Wi-Fi Standards

Wi-Fi Gen	IEEE Standard	Release Date	2.4 GHz	5 GHz	Max Data Rate
Wi-Fi	802.11	1997	Yes	No	2 Mbps
Wi-Fi 1	802.11b	1999	Yes	No	11 Mbps
Wi-Fi 2	802.11a	1999	No	Yes	54 Mbps
Wi-Fi 3	802.11g	2003	Yes	No	54 Mbps
Wi-Fi 4	802.11n	2009	Yes	Yes	600 Mbps
Wi-Fi 5	802.11ac	2013	No	Yes	6.93 Gbps
Wi-Fi 6	802.11ax	2019	Yes	Yes	14 Gbps

IOT Data link layer and Network layer protocols

IEEE 802.11



IOT Data link layer and Network layer protocols

IEEE 802.11

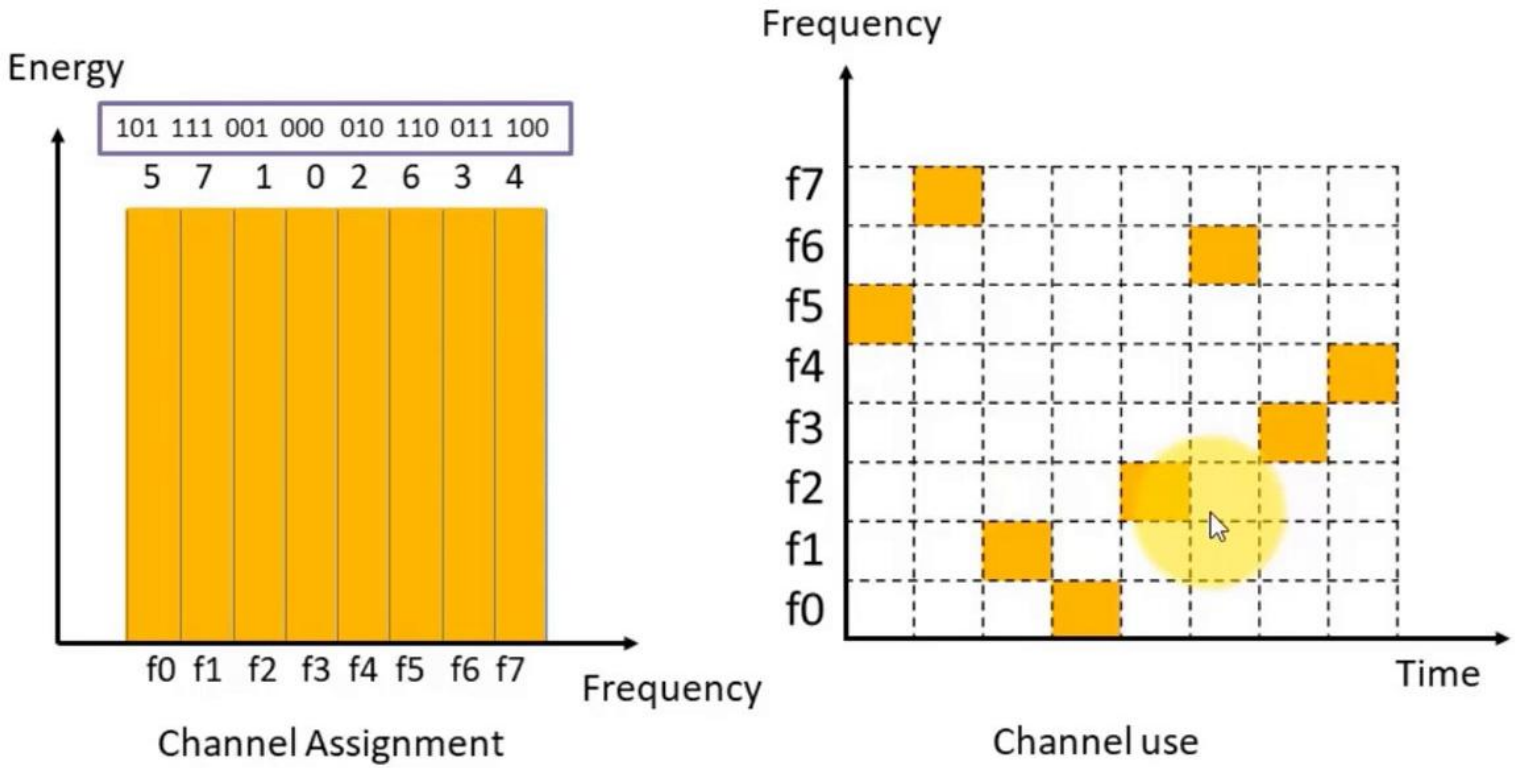
- IEEE 802.11:
- was the original version released in 1997.
- It provided 1 Mbps or 2 Mbps data rate in the 2.4 GHz band and used either frequency-hopping spread spectrum (FHSS) or direct-sequence spread spectrum (DSSS).
- It is obsolete now.



IOT Data link layer and Network layer protocols

IEEE 802.11

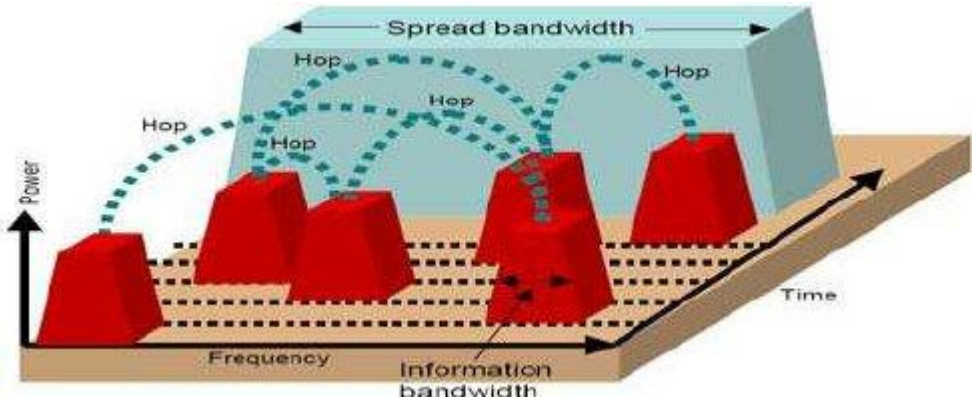
Frequency-hopping Spread Spectrum (FHSS)

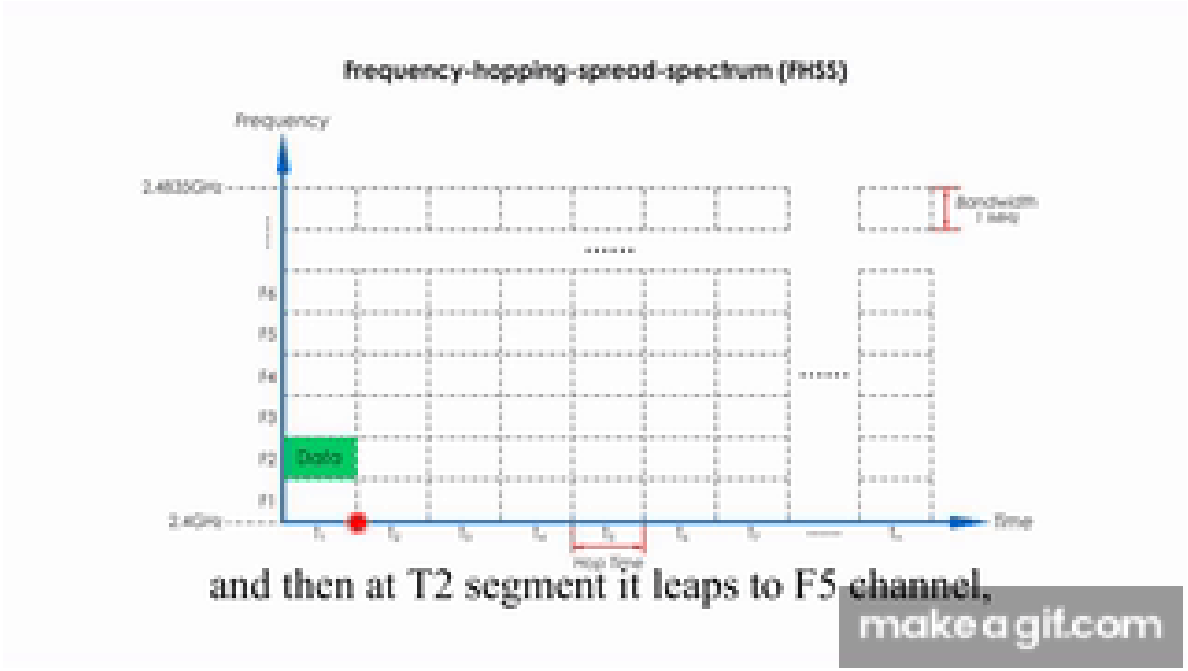


IOT Data link layer and Network layer protocols

IEEE 802.11

Frequency-hopping Spread Spectrum (FHSS)

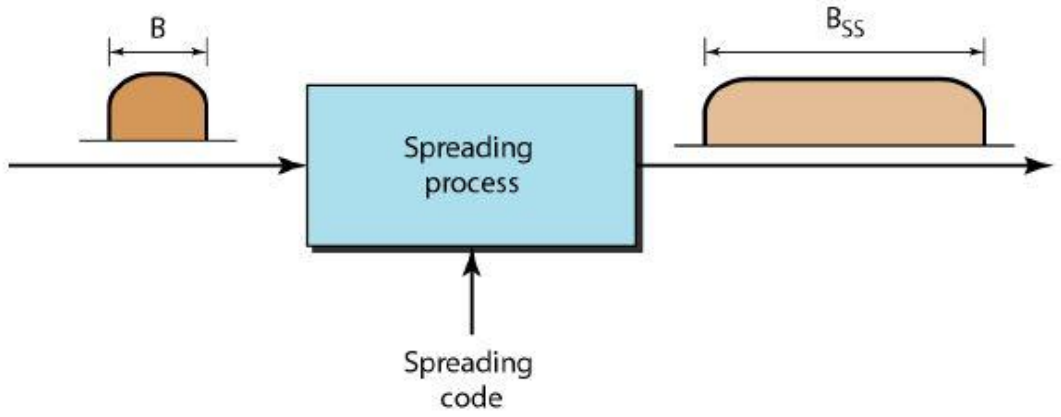


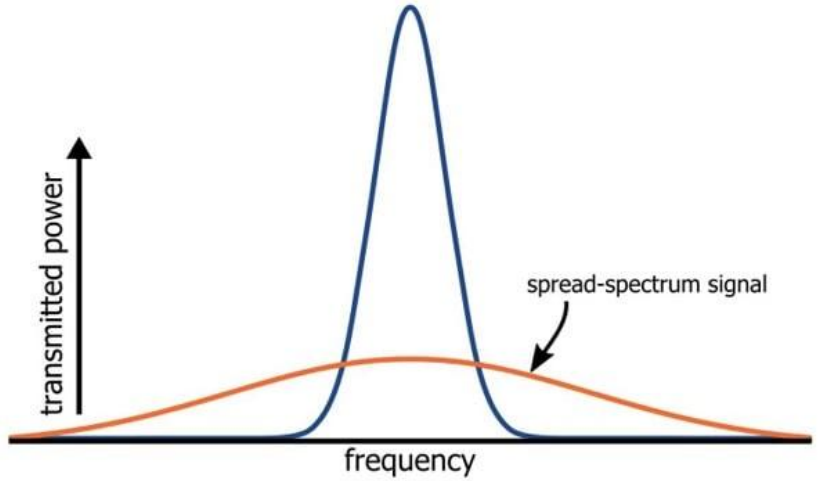


IOT Data link layer and Network layer protocols

IEEE 802.11

Direct-sequence Spread Spectrum (DSSS)





The spreading code is a series of numbers that look random, but are actually a pattern

IOT Data link layer and Network layer protocols

IEEE 802.11

- IEEE 802.11a :
- was published in 1999 as a modification to 802.11
- with orthogonal frequency division multiplexing (OFDM) based air interface in physical layer instead of FHSS or DSSS of 802.11.
- It provides a maximum data rate of 54 Mbps operating in the 5 GHz band.
- Besides it provides error correcting code.
- As 2.4 GHz band is crowded, relatively sparsely used 5 GHz imparts additional advantage to 802.11a.
- Further amendments to 802.11a are 802.11ac, 802.11ad, 802.11af, 802.11ah, 802.11ai, 802.11aj etc

IOT Data link layer and Network layer protocols

IEEE 802.11

- IEEE 802.11b:
- Is a direct extension of the original 802.11 standard that appeared in early 2000.
- It uses the same modulation technique as 802.11, i.e. DSSS and operates in the 2.4 GHz band.
- It has a higher data rate of 11 Mbps as compared to 2 Mbps of 802.11, due to which it was rapidly adopted in wireless LANs.
- However, since 2.4 GHz band is pretty crowded, 802.11b devices faces interference from other devices.
- Further amendments to 802.11b are 802.11ba, 802.11bb, 802.11bc, 802.11bd and 802.11be

IOT Data link layer and Network layer protocols

IEEE 802.11

- IEEE 802.11g :
- was indorsed in 2003.
- It operates in the 2.4 GHz band (as in 802.11b) and provides a average throughput of 22 Mbps.
- It uses OFDM technique (as in 802.11a).
- It is fully backward compatible with 802.11b.
- 802.11g devices also faces interference from other devices operating in 2.4 GHz band.

IOT Data link layer and Network layer protocols

IEEE 802.11

- IEEE 802.11n:
- was approved and published in 2009 that operates on both the 2.4 GHz and the 5 GHz bands.
- It has variable data rate ranging from 54 Mbps to 600 Mbps.
- It provides a marked improvement over previous standards 802.11 by incorporating multiple-input multiple-output antennas (MIMO antennas).

IOT Data link layer and Network layer protocols

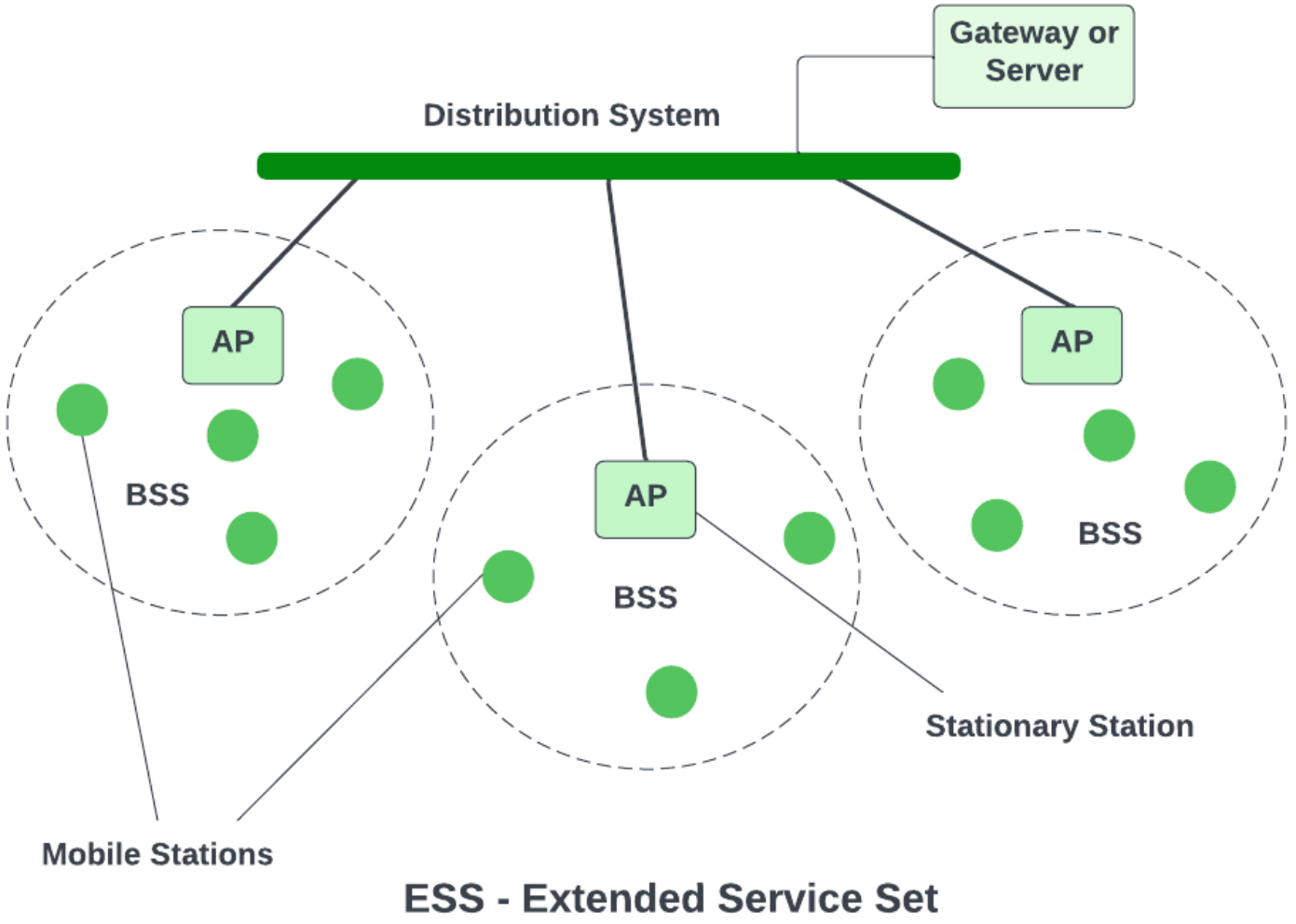
IEEE 802.11

- IEEE 802.11p:
- Is an amendment for including wireless access in vehicular environments (WAVE) to support Intelligent Transportation Systems (ITS).
- They include network communications between vehicles moving at high speed and the environment.
- They have a data rate of 27 Mbps and operate in 5.9 GHz band.

IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:



IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:

- Wireless LANs are those Local Area Networks that use high frequency radio waves instead of cables for connecting the devices in LAN.
- Users connected by WLANs can move around within the area of network coverage.
- Most WLANs are based upon the standard IEEE 802.11 or WiFi.

IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:

- The components of IEEE 802.11 architecture

i. Stations (STA)

ii. Basic Service Set (BSS)

iii. Extended Service Set (ESS)

iv. Distribution System (DS)

IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:

i. Stations (STA):

- Stations comprise all devices and equipments that are connected to the wireless LAN.
- A station can be of two types:
  - Wireless Access Pointz (WAP) – WAPs or simply access points (AP) are generally wireless routers that form the base stations or access.
  - Client. – Clients are workstations, computers, laptops, printers, smart phones, etc.
- Each station has a wireless network interface controller.

IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:

ii. Basic Service Set (BSS)

- A basic service set is a group of stations communicating at physical layer level.
- BSS can be of two categories depending upon mode of operation:
  - Infrastructure BSS – Here, the devices communicate with other devices through access points.
  - Independent BSS – Here, the devices communicate in peer-to-peer basis in an ad hoc manner.

IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:

iii. Extended Service Set (ESS)

- It is a set of all connected BSS.

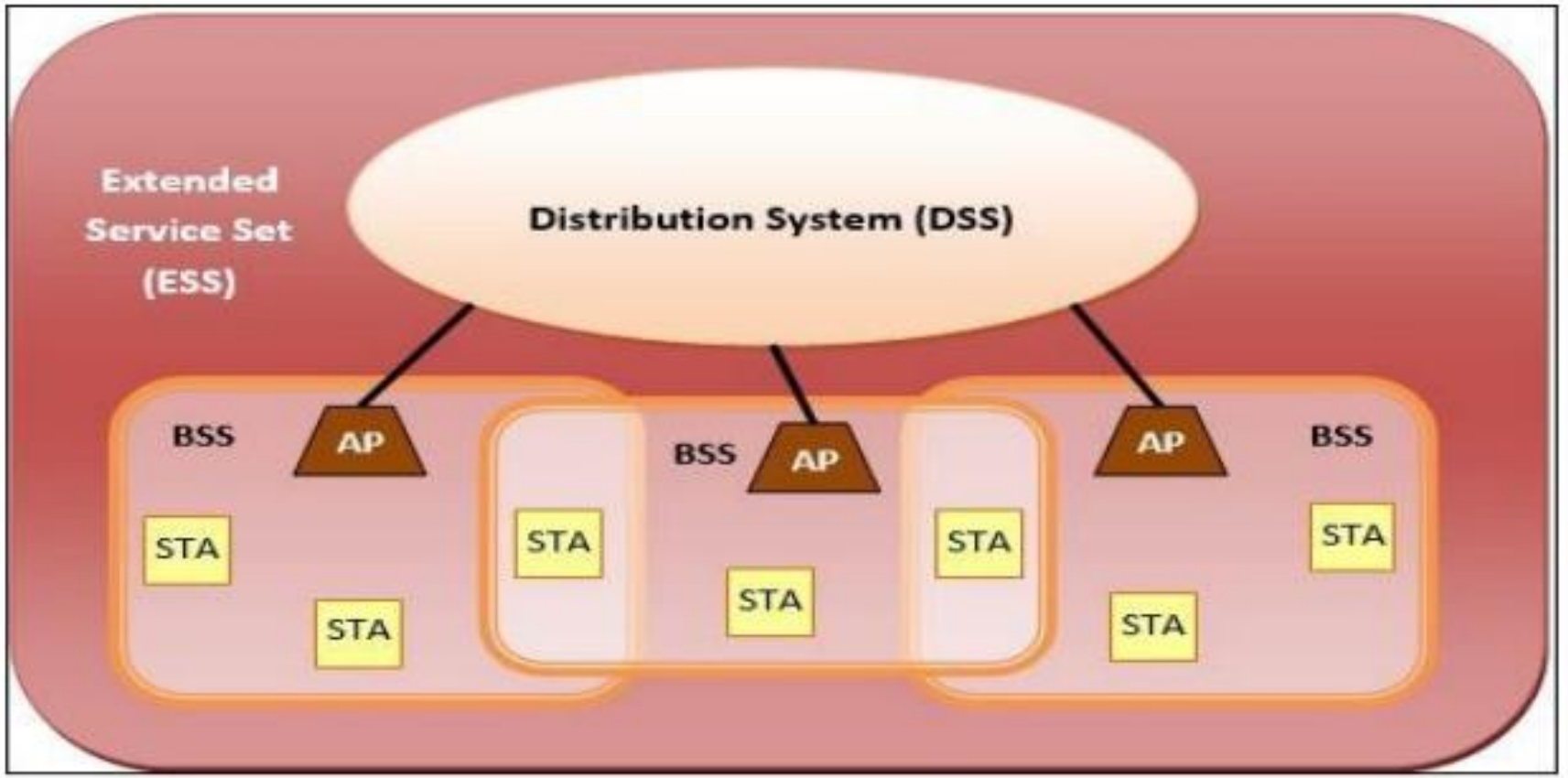
iv. Distribution System (DS)

- It connects access points in ESS.

IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:



IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:

- An ESS can also provide gateway access for wireless users into a wired network.
- Each end station associates itself with one access point.
- If station A, associated with AP-1 wants to send a frame to another station associated with AP- 2, the first sends a frame to its access point (AP-1), which forwards the frame across the distribution system to the access point AP-2.
- AP-2 finally delivers it to the destination station.

IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:

- The technique used for this purpose is known as scanning, which involves the following steps:
- A station sends a probe frame.
- All APs within reach reply with a probe response frame.
- The station selects one of the access points, and sends the AP an Association Request frame.
- The AP replies with an Association Response frame.



IOT Data link layer and Network layer protocols

IEEE 802.11

- Medium Access Control:

  - CSMA CA stands for Carrier Sense Multiple Access – Collison Avoidance.
  - It is a network protocol that operates in the data link layer of the OSI model.
  - It uses carrier sensing to check if the channel is idle before transmitting data.

IOT Data link layer and Network layer protocols

IEEE 802.11

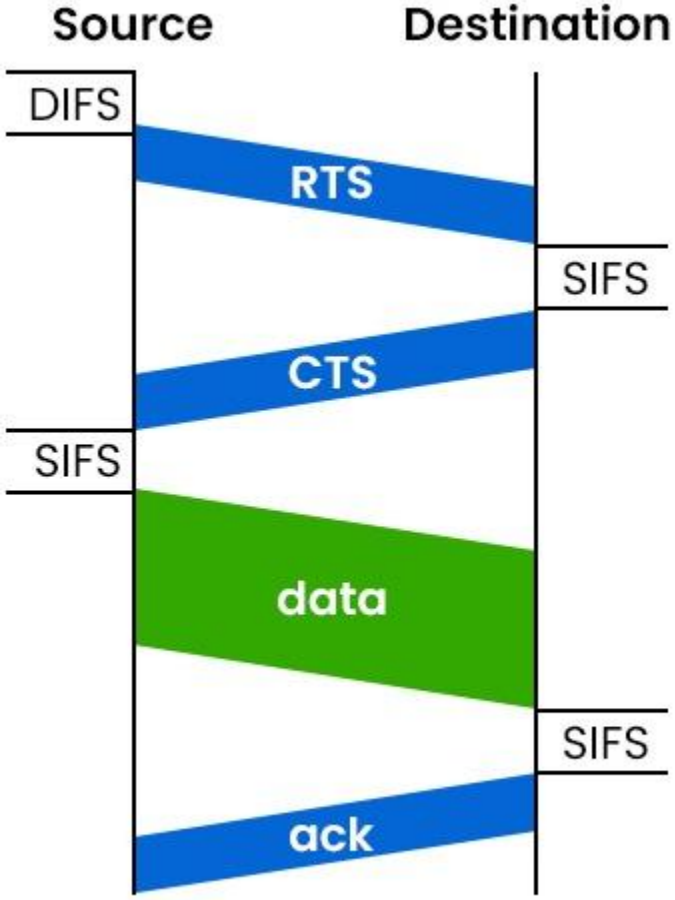
- Medium Access Control:
  - It also uses collision avoidance techniques such as interframe spacing, contention windows, and acknowledgments to reduce the chances of collisions.
  - CSMA/CA is mainly used in wireless networks, where collision detection is difficult or impossible.



IOT Data link layer and Network layer protocols

IEEE 802.11

- Medium Access Control:
  - CSMA CA
    - To prevent a collision, devices rely on signals called "Ready to Send" (RTS) and "Clear to Send" (CTS).
    - A device will scan the airwaves for a signal when it is ready to send.
    - If there isn't already one, it will create a request to transmit (RTS) signal.
    - It then delivers its own data and concludes with a CTS signal, which lets other wireless devices know it's their turn to communicate.



CSMA CA

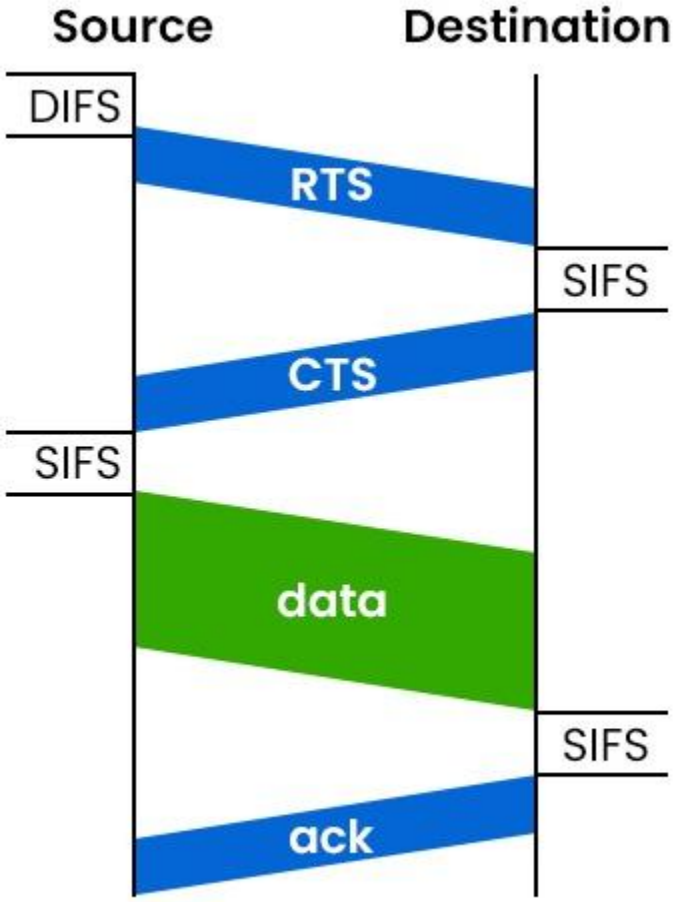
IOT Data link layer and Network layer protocols

IEEE 802.11

- Medium Access Control:
  - CSMA CA

DIFS- distributed inter-frame space- is the time a wireless station waits to send data after checking that the channel is clear

SIFS-Short Interframe Space-SIFS is the smallest IFS and is used for sending an ACK after receiving data.



CSMA CA

IOT Data link layer and Network layer protocols

IEEE 802.11

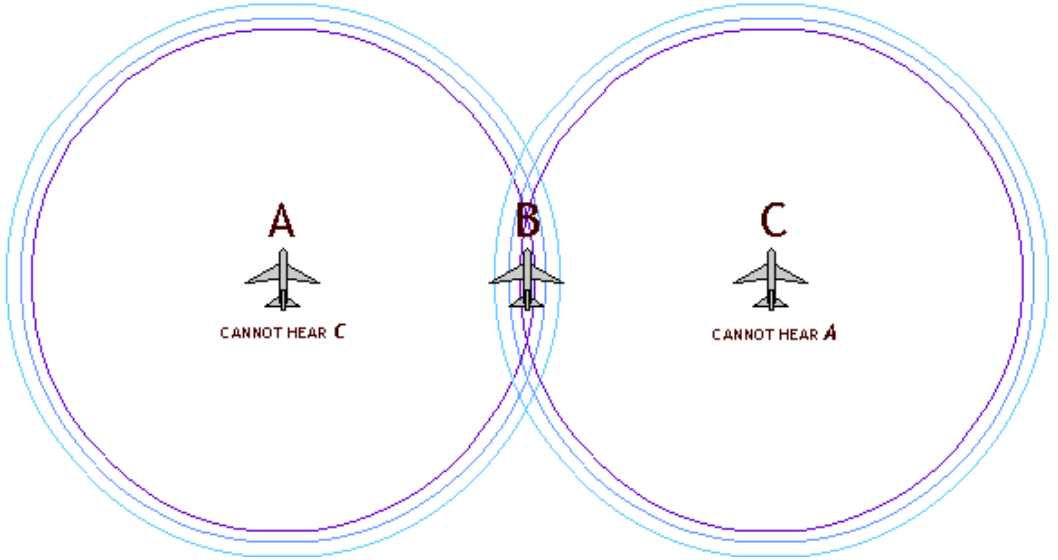


- Medium Access Control:
  - However, the wireless medium presents some unique challenges not present in wired LANs that must be dealt with by the MAC used for IEEE 802.11.
  - Some of the challenges are:
    - The wireless LAN is prone to more interference and is less reliable.
    - The wireless LAN is susceptible to unwanted interception leading to security problems.
    - There are so called hidden station and exposed station problems.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Medium Access Control:
  - The Hidden Station Problem
    - A is transmitting to B
    - If C senses the media, it will not hear anything because it is out of range
    - thus will falsely conclude that no transmission is going on and will start transmit to B.
    - the transmission will interfere at B, wiping out the frame from A
    - C act as a hidden station to A

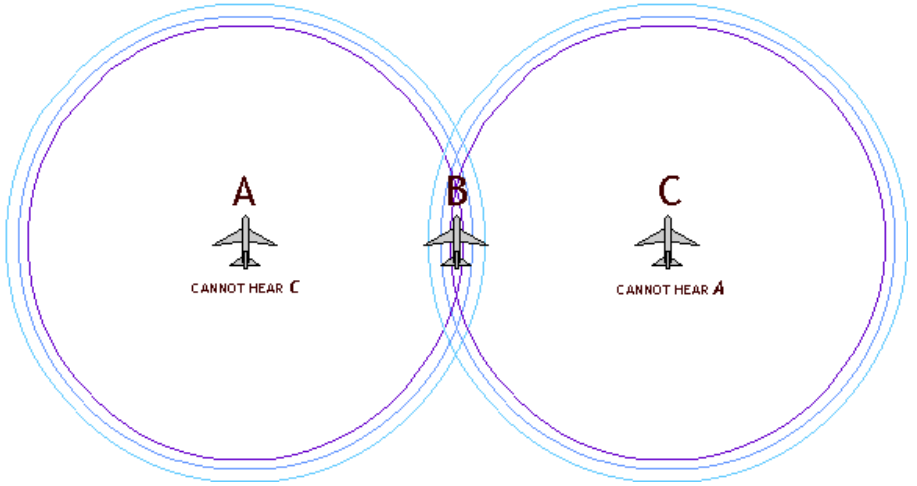


IOT Data link layer and Network layer protocols

IEEE 802.11

- Medium Access Control:

  - The Hidden Station Problem

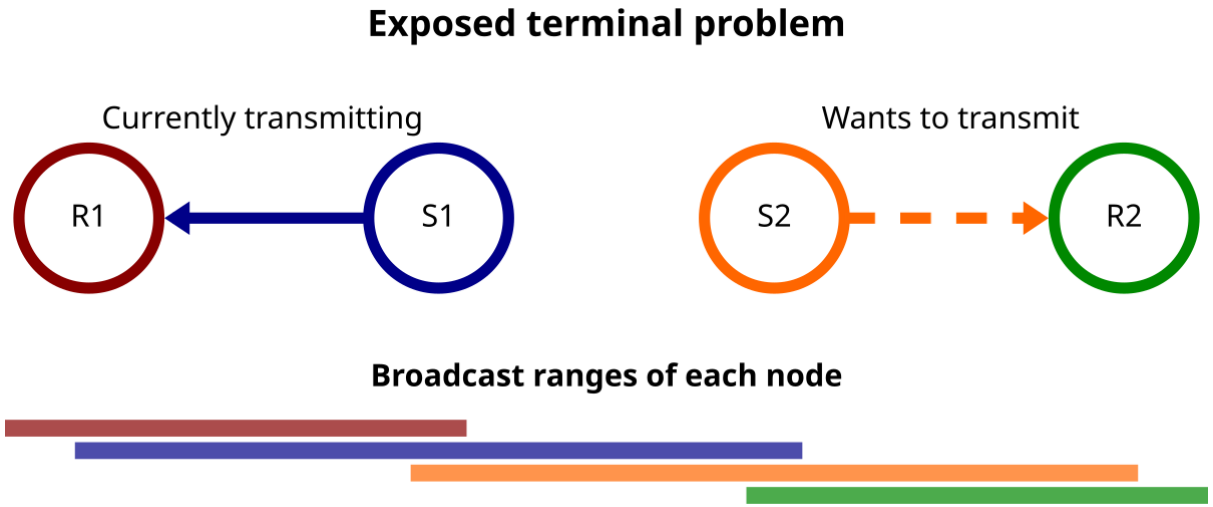


  - The problem of a station not been able to detect a potential competitor for the medium because the competitor is too far away is referred as Hidden Station Problem.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Medium Access Control:
  - Exposed Station problem

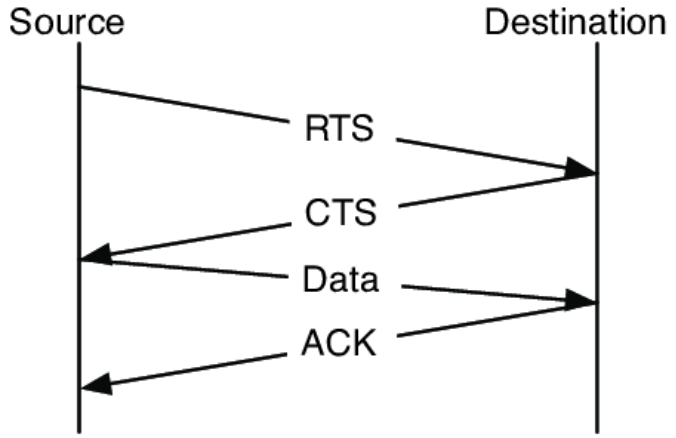


- S1 is transmitting to R1,
- and S2 sense the medium and detects the ongoing transmission between S1 and R1.
- S2 falsely conclude that it cannot transmit to R2, when the fact is that such transmission would cause no problem.
- A transmission could cause a problem only when the destination is in zone between S1 and S2.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Remedy to Hidden Node and Exposed node Problem



•**Use request-to-send/clear-to-send (RTS/CTS) handshaking**:

•This is implemented at the Access Point along with the Carrier sense multiple access with collision avoidance (CSMA/CA) scheme.

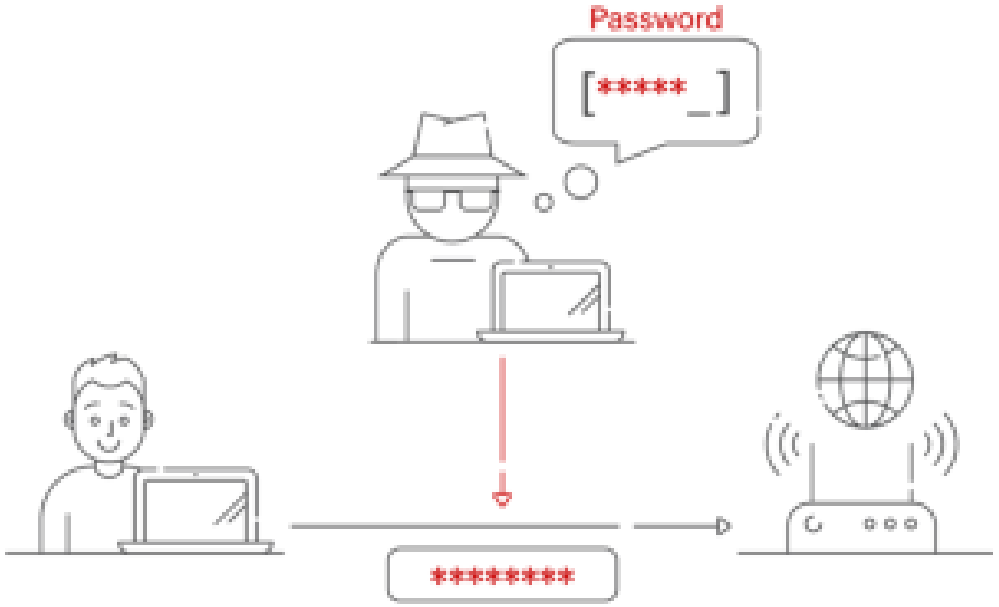
• RTS/CTS handshaking helps to overcome the hidden node problem, but it's not a complete solution.



IOT Data link layer and Network layer protocols

IEEE 802.11

- Security in MAC layer

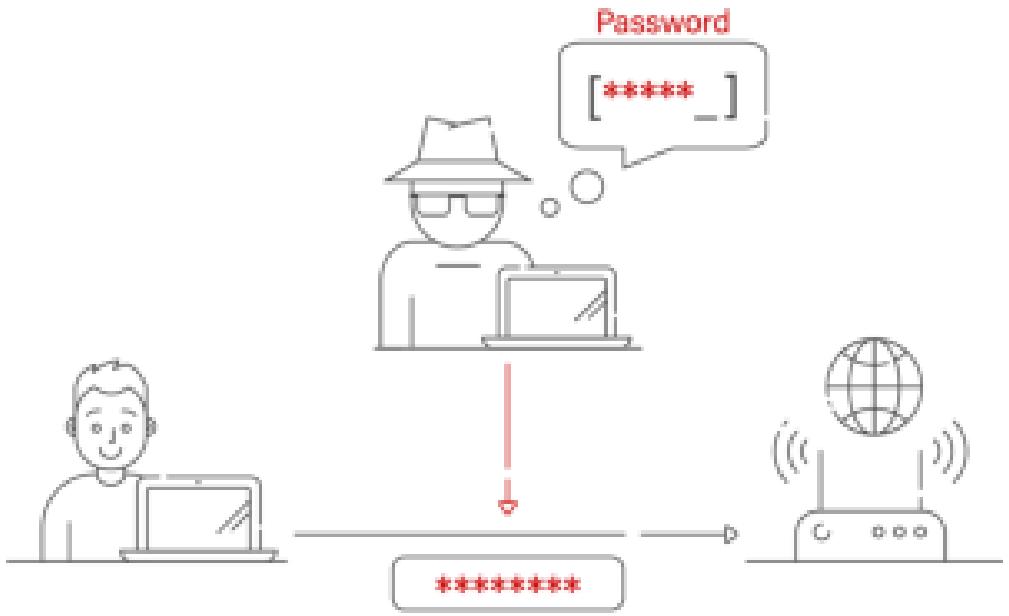


- Wireless LANs are subjected to possible breaches from unwanted monitoring.
- To overcome this problem, IEEE 802.11 specifies an optional MAC layer security system known as Wired Equivalent Privacy (WEP).

IOT Data link layer and Network layer protocols

IEEE 802.11

- Security in MAC layer
  - The objective is to provide a level of privacy to the wireless LAN similar to that enjoyed by wired Ethernets.
  - It is achieved with the help of a 40-bit shared key authentication service.

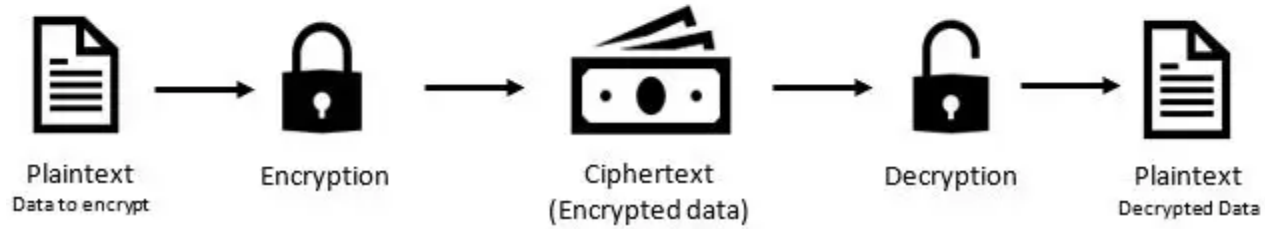


IOT Data link layer and Network layer protocols

IEEE 802.11

- Security in MAC layer
  - By default each BSS supports up to four 40-bit keys that are shared by all the clients in the BSS.
  - Keys unique to a pair of communicating clients and direction of transmission may also be used.
  - Advanced Encryption Standard (AES) for authentication and encryption is recommended as a long term solution.

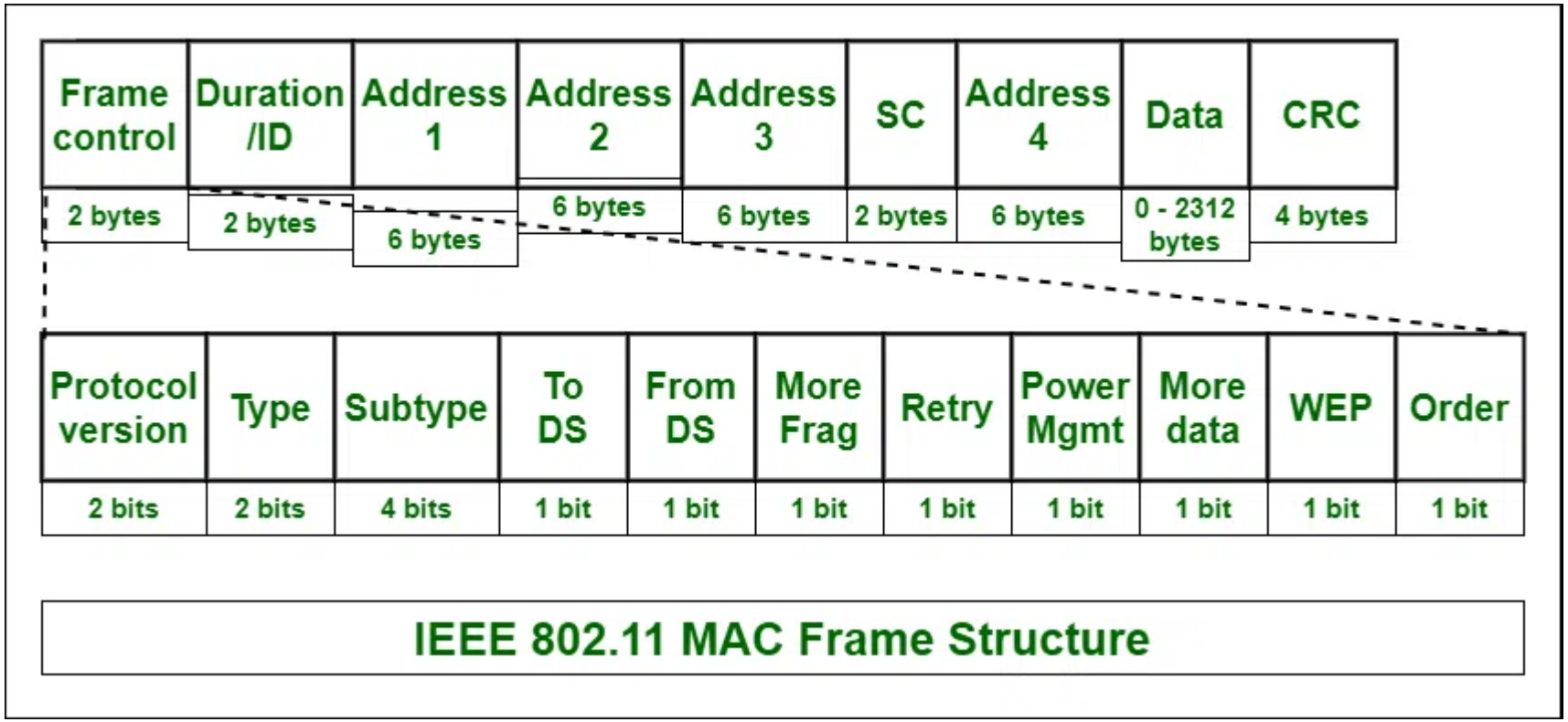




IOT Data link layer and Network layer protocols

IEEE 802.11

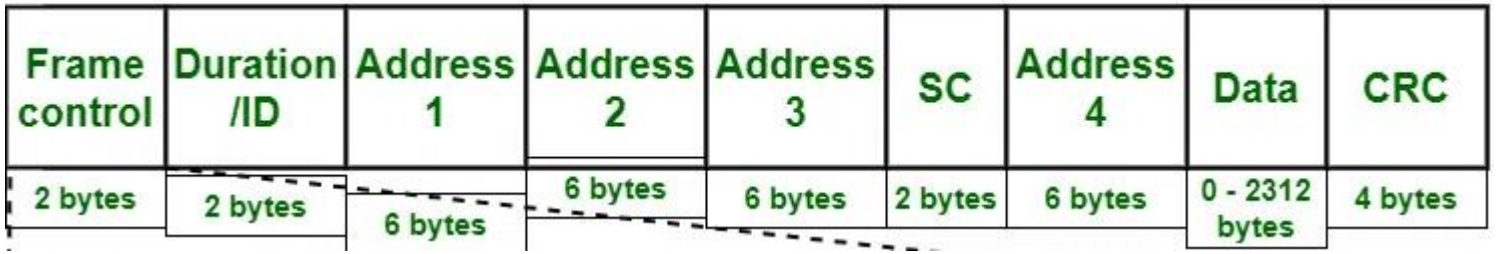
- Frame Format of IEEE 802.11



IOT Data link layer and Network layer protocols

IEEE 802.11

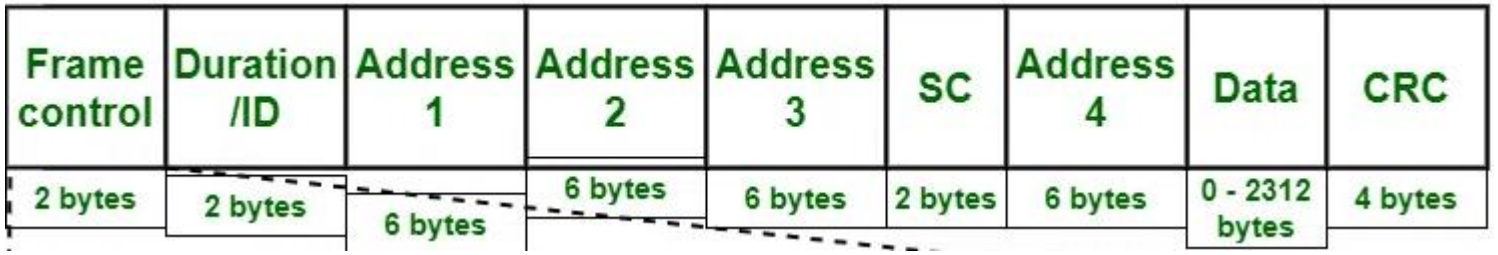
- Frame Format of IEEE 802.11



- The main fields of a frame of wireless LANs as laid down by IEEE 802.11 are
- Frame Control –
  - It is a 2 bytes starting field composed of 11 subfields.
  - It contains control information of the frame.
- Duration –
  - It is a 2-byte field that specifies the time period for which the frame and its acknowledgment occupy the channel.

IOT Data link layer and Network layer protocols

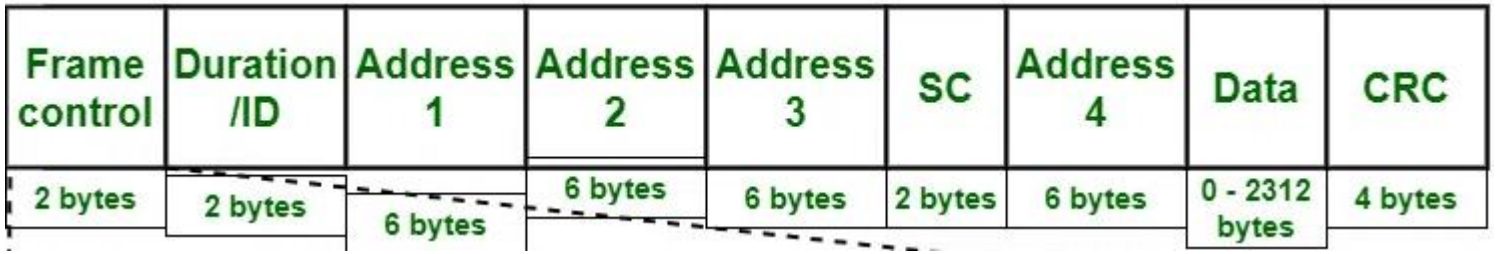
IEEE 802.11



- Frame Format of IEEE 802.11
  - Address fields –
    - There are three 6-byte address fields containing addresses of source, immediate destination, and final endpoint respectively.
  - Sequence –
    - It a 2 bytes field that stores the frame numbers.

IOT Data link layer and Network layer protocols

IEEE 802.11



- Frame Format of IEEE 802.11
  - Data –
    - This is a variable-sized field that carries the data from the upper layers.
    - The maximum size of the data field is 2312 bytes.
  - Check Sequence –
    - It is a 4-byte field containing error detection information.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Frame Format of IEEE 802.11

Protocol version	Type	Subtype	To DS	From DS	More Frag	Retry	Power Mgmt	More data	WEP	Order
2 bits	2 bits	4 bits	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit

Frame Control(FC) –

It is 2 bytes long field which defines type of frame and some control information. Various fields presentation FC are:

1. Version: It is a 2 bit long field which indicates the current protocol version which is fixed to be 0 for now.

2. Type: It is a 2 bit long field which determines the function of frame i.e management (00), control (01) or data (10). The value 11 is reserved.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Frame Format of IEEE 802.11

Protocol version	Type	Subtype	To DS	From DS	More Frag	Retry	Power Mgmt	More data	WEP	Order
2 bits	2 bits	4 bits	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit

Frame Control(FC) –

3. Subtype: It is a 4 bit long field which indicates sub-type of the frame like 0000 for association request, 1000 for beacon.

4. To DS: It is a 1 bit long field which when set indicates that destination frame is for DS(distribution system).

5. From DS: It is a 1 bit long field which when set indicates frame coming from DS.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Frame Format of IEEE 802.11

Protocol version	Type	Subtype	To DS	From DS	More Frag	Retry	Power Mgmt	More data	WEP	Order
2 bits	2 bits	4 bits	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit

Frame Control(FC) –

6. More frag (More fragments): It is 1 bit long field which when set to 1 means frame is followed by other fragments.

7. Retry: It is 1-bit long field, if the current frame is a retransmission of an earlier frame, this bit is set to 1.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Frame Format of IEEE 802.11

Protocol version	Type	Subtype	To DS	From DS	More Frag	Retry	Power Mgmt	More data	WEP	Order
2 bits	2 bits	4 bits	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit

Frame Control(FC) –

8. Power Mgmt (Power management): It is 1-bit long field that indicates the mode of a station after successful transmission of a frame. Set to 1 the field indicates that the station goes into power-save mode. If the field is set to 0, the station stays active.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Frame Format of IEEE 802.11

Protocol version	Type	Subtype	To DS	From DS	More Frag	Retry	Power Mgmt	More data	WEP	Order
2 bits	2 bits	4 bits	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit

Frame Control(FC) –

9. More data: It is 1-bit long field that is used to indicate receiver that a sender has more data to send than the current frame.

This can be used by an access point to indicate to a station in power-save mode that more packets are buffered or it can be used by a station to indicate to an access point after being polled that more polling is necessary as the station has more data ready to transmit.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Frame Format of IEEE 802.11

Protocol version	Type	Subtype	To DS	From DS	More Frag	Retry	Power Mgmt	More data	WEP	Order
2 bits	2 bits	4 bits	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit

Frame Control(FC) –

10. WEP: It is 1 bit long field which indicates that the standard security mechanism of 802.11 is applied.

11. Order: It is 1 bit long field, if this bit is set to 1 the received frames must be processed in strict order.

IOT Data link layer and Network layer protocols

IEEE 802.11



ESP-01 ESP8266 Serial WiFi Transceiver Module

IOT Data link layer and Network layer protocols

IEEE 802.15

What is



wireless specialty networks (WSNs)-

WSNs include

- wireless personal area networks (WPANs),
- Bluetooth,
- mesh networks, and
- body area networks.

IOT Data link layer and Network layer protocols

IEEE 802.15

- IEEE 802.15.1: WPAN / Bluetooth
- IEEE 802.15.2: Coexistence
- IEEE 802.15.3: High Rate WPAN
- IEEE 802.15.4: Low Rate WPAN
- IEEE 802.15.5: Mesh Networking
- IEEE 802.15.6: Body Area Networks
- IEEE 802.15.7: Visible Light Communication
- IEEE P802.15.8: Peer Aware Communications
- IEEE P802.15.9: Key Management Protocol
- IEEE P802.15.10: Layer 2 Routing
- IEEE 802.15.13: Multi-Gigabit/s Optical Wireless Communications

IOT Data link layer and Network layer protocols

IEEE 802.15

IEEE 802.15.1



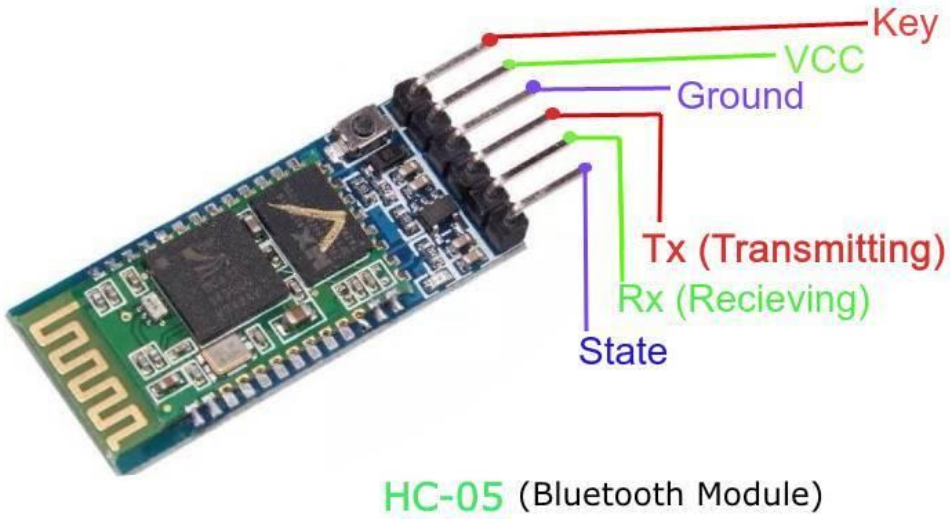
- **IEEE definition of WPAN**

- Wireless personal area networks (WPANs) are used to convey information over short distances among a private, intimate group of participant devices.

IOT Data link layer and Network layer protocols

IEEE 802.15

IEEE 802.15.1



HC-05 (Bluetooth Module)

- **WPAN**
- Unlike a wireless local area network (WLAN), a connection made through a WPAN involves little or no infrastructure or direct connectivity to the world outside the link.
- This allows small, power-efficient, inexpensive solutions to be implemented for a wide range of devices.

IOT Data link layer and Network layer protocols

IEEE 802.15



- **IEEE 802.15.1: WPAN / Bluetooth**

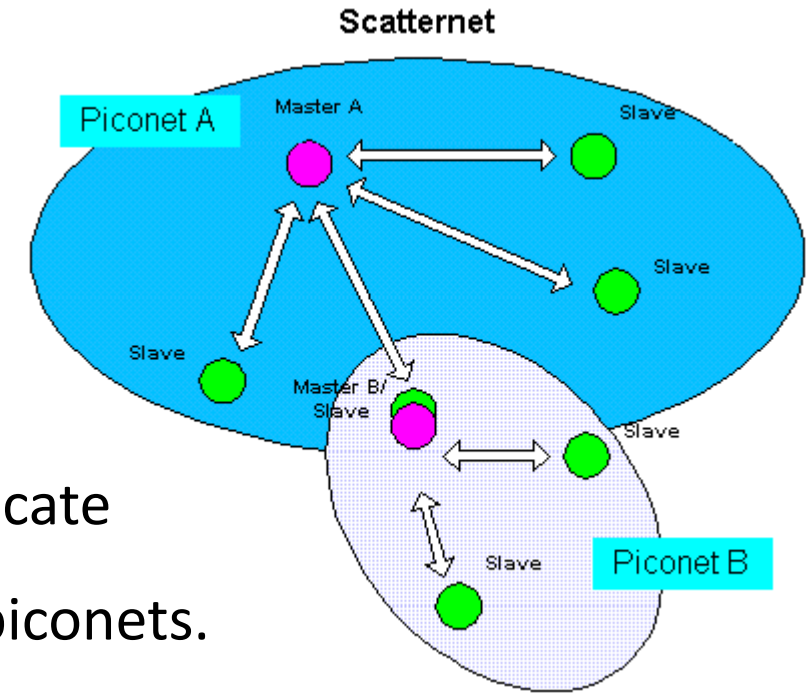
- A widely used WPAN technology is known as Bluetooth (version 1.2 or version 2.0)
- The IEEE 802.15.1 standard specifies the architecture and operation of Bluetooth devices, but only as far as physical layer and media access control (MAC) layer operation is concerned (the core system architecture).
- Higher protocol layers and applications defined in usage profiles are standardized by the Bluetooth SIG.

•In 1998, Ericsson, IBM, Intel, Nokia and Toshiba formed the "Bluetooth Special Interest Group" to establish standards for short-range radio.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.1: WPAN / Bluetooth**



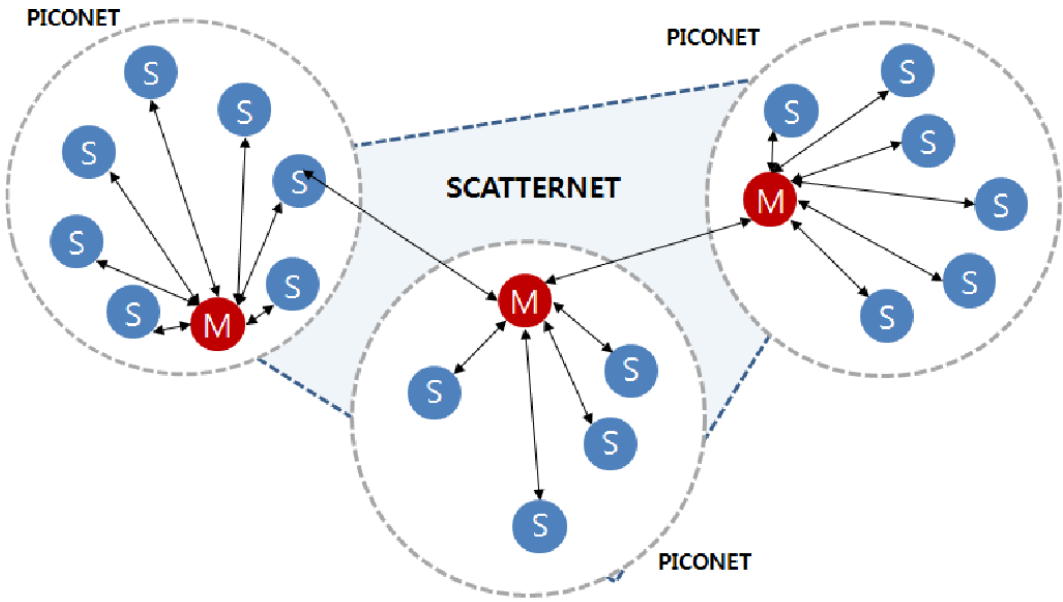
- Piconets
- Bluetooth enabled electronic devices connect and communicate wirelessly through short-range, ad hoc networks known as piconets.
- Piconets are established dynamically and automatically as Bluetooth enabled devices enter and leave radio proximity.

Up to 8 devices in one piconet (1 master and 7 slave devices). Max range 10 m.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.1: WPAN / Bluetooth**



- Scatternet
- It is a network which connects multiple piconets using Bluetooth and it acts as a master and another type of piconet acts as a slave.
- It has more than 6 devices that can be connected.
- Multiple devices are active, so there is an effective use of channel bandwidth.
- It is a connection of multiple piconets therefore it is applicable for devices belonging to large areas.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.1: WPAN / Bluetooth**

Specifications

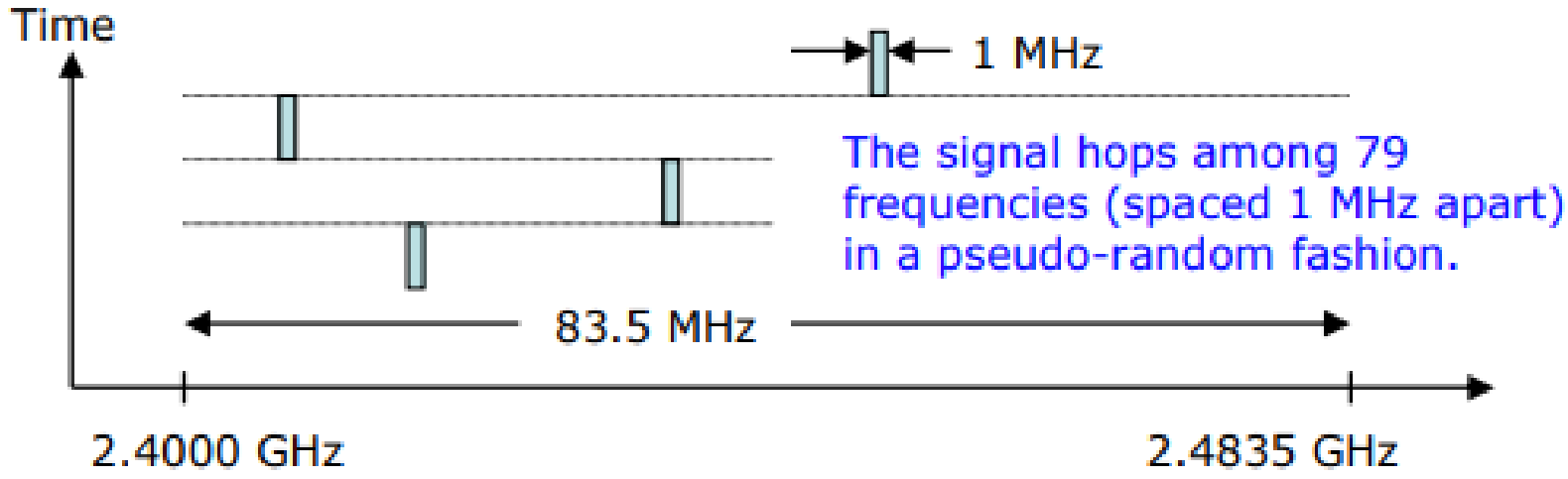
Topology	Up to 7 simultaneous links
Modulation	Gaussian filtered FSK
RF bandwidth	220 kHz (-3 dB), 1 MHz (-20 dB)
RF band	2.4 GHz ISM frequency band
RF carriers	79 (23 as reduced option)
Carrier spacing	1 MHz
Access method	FHSS-TDD-TDMA
Freq. hop rate	1600 hops/s

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.1: WPAN / Bluetooth**

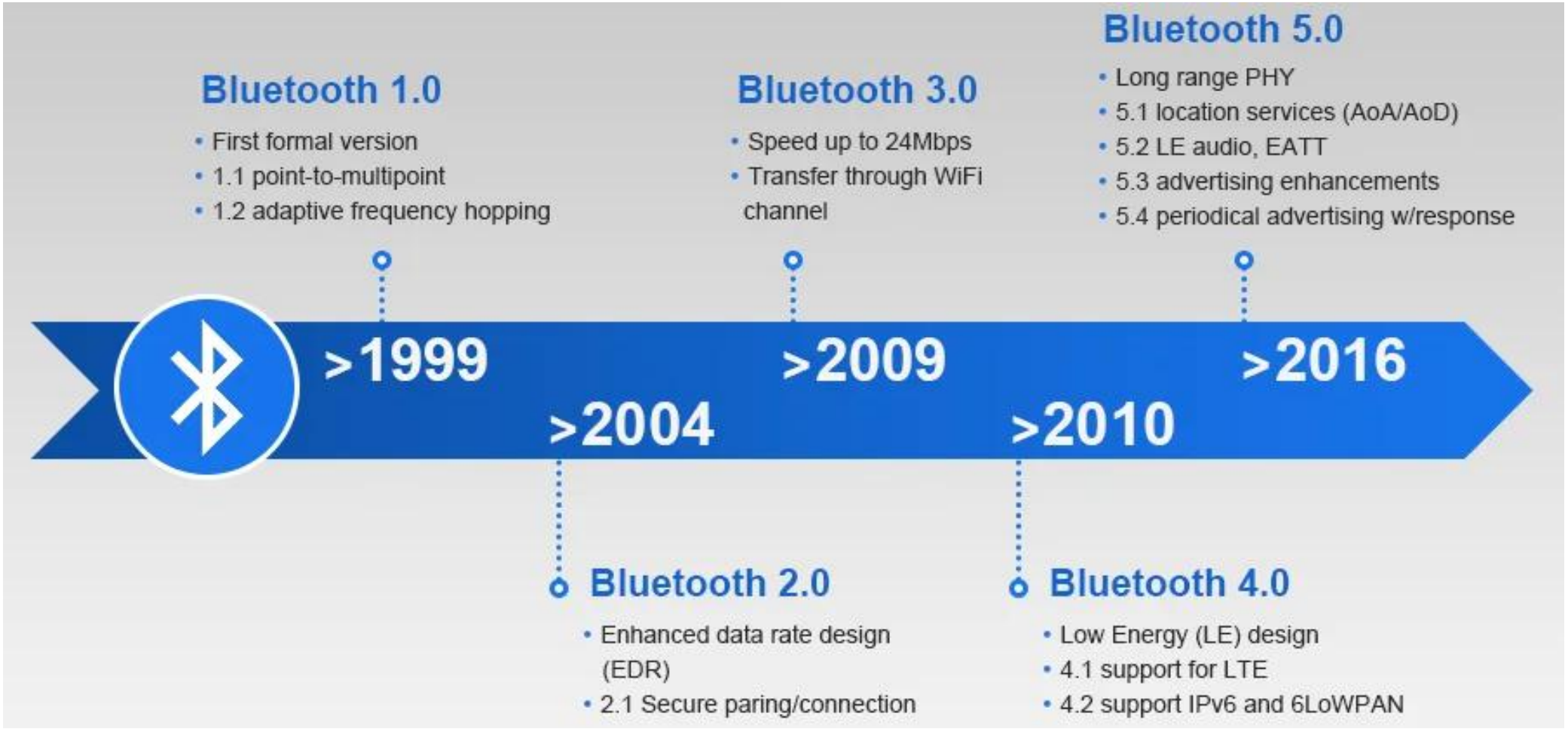
Bluetooth technology operates in the 2.4 GHz ISM band, using a spread spectrum, frequency hopping, full-duplex signal at a nominal rate of 1600 hops/second.



IOT Data link layer and Network layer protocols

IEEE 802.15

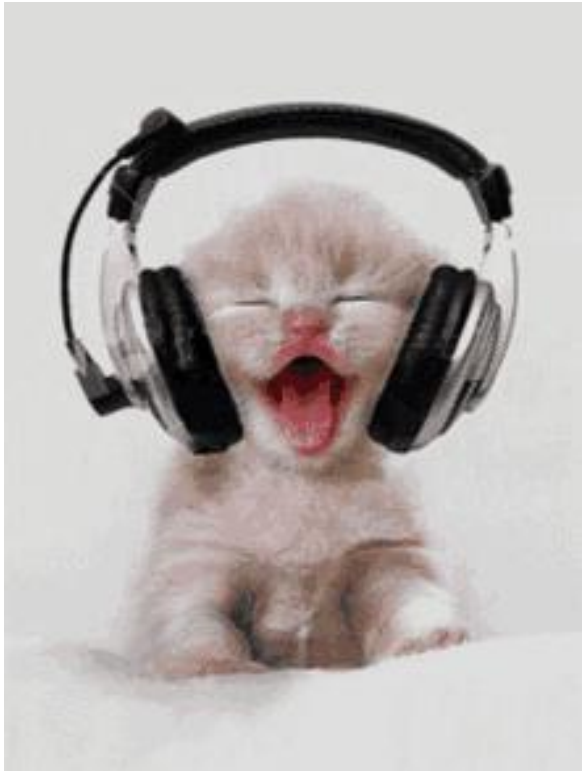
- **IEEE 802.15.1: WPAN / Bluetooth**



IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.1: WPAN / Bluetooth**
  - **Types**
  - **Classic Bluetooth versions: Bluetooth 1.0 – 3.0**
  - **Bluetooth low energy versions: Bluetooth 4.0 – 5.4**



IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.1: WPAN / Bluetooth**
  - **Bluetooth low energy versions: Bluetooth 4.0 – 5.4**







IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.1: WPAN / Bluetooth**

	Bluetooth® Classic	Bluetooth® Low Energy
Frequency Band	2.4 GHZ ISM Band	2.4 GHZ ISM Band
No. of Channels	79 one MHz Channel	40 two MHz Channel
Power Consumption	Low	Less
Data Rate	1-3 Mbps	1 Mbps
Latency	Approx. 100 ms	Approx. 6 ms
Range	< 30 m	50 m (150 m in open area)

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.1: WPAN / Bluetooth**

	Bluetooth® Classic	Bluetooth® Low Energy
Topology	Peer-to-peer (1:1)	Peer-to-peer (1:1) Star (many:1) Broadcast (1:many) Mesh (many:many)
Device pairing	Required	Not Required
Voice capable	Yes	No

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.1: WPAN / Bluetooth**

	Bluetooth® Classic	Bluetooth® Low Energy
Nodes/Active Slaves	7	Unlimited
Security	64b/128 bit, user defined application layer	128 bits AES, user defined application layer
Smartphone Compatibility	100% available on smartphones	100% available on smartphones
Use Cases	Streaming applications like audio streaming, file transfer, and headsets	Location beacons, smart home applications, medical devices, industrial monitoring, fitness trackers

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.1: WPAN / Bluetooth**

Classic Bluetooth versions: Bluetooth 1.0 – 3.0

Version	Year of release	Max transmission range	Max range
Bluetooth 1.0	1999	732.2 kbit/s	10 m (33 ft)
Bluetooth 1.1	2001	732.2 kbit/s	10 m (33 ft)
Bluetooth 1.2	2003	1 Mbps	10 m (33 ft)
Bluetooth 2.0	2004	2.1 Mbps	30 m (100 ft)
Bluetooth 2.1	2007	2.1 Mbps	30 m (100 ft)
Bluetooth 3.0	2009	24 Mbps	30 m (100 ft)

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.1: WPAN / Bluetooth**

  Bluetooth low energy versions: Bluetooth 4.0 – 5.4

Version	Release Year	Max transmission speed	Max range
Bluetooth 4.0	2009	1 Mbps (LE) 3 Mbps (EDR)	60 m (200 ft)
Bluetooth 4.1	2013	1 Mbps (LE) 3 Mbps (EDR)	60 m (200 ft)
Bluetooth 4.2	2014	1 Mbps (LE) 3 Mbps (EDR)	60 m (200 ft)

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.1: WPAN / Bluetooth**

  Bluetooth low energy versions: Bluetooth 4.0 – 5.4

Version	Release Year	Max transmission speed	Max range
Bluetooth 5.0	2016	2 Mbps (LE) 50 Mbps (EDR)	240 m (800 ft)
Bluetooth 5.1	2019	2 Mbps (LE) 50 Mbps (EDR)	240 m (800 ft)
Bluetooth 5.2	2020	2 Mbps (LE) 50 Mbps (EDR)	240 m (800 ft)
Bluetooth 5.3	2021	2 Mbps (LE) 50 Mbps (EDR)	240 m (800 ft)
Bluetooth 5.4	2023	2 Mbps (LE) 50 Mbps (EDR)	240 m (800 ft)

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.2: Coexistence**
  - IEEE 802.15.2 is a recommended practice that describes how wireless personal area networks (WPANs) and wireless local area networks (WLANs) can coexist
  - The standard's goal is to enable WPAN devices to operate alongside other wireless devices in unlicensed frequency bands

The standard was published in 2003, reaffirmed in 2009, and withdrawn in 2018

IEEE 802.15.3 — Overview

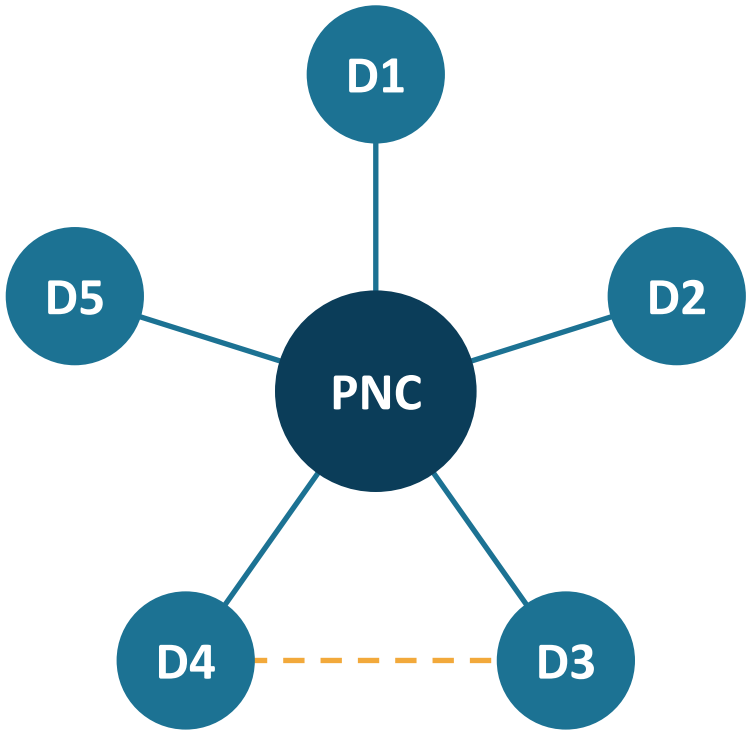
11–55	~10 m	2003	2.4 GHz
Mbps data rate	operating range	year ratified	ISM band

Why 802.15.3 exists

- Bluetooth (802.15.1) was too slow for multimedia; Wi-Fi (802.11) consumed too much power for handheld devices.
- 802.15.3 fills the gap: high throughput plus low power, aimed at streaming audio/video between nearby consumer devices.
- Forms self-organizing, ad hoc networks called piconets with no fixed infrastructure required.

802.15.3 Network Architecture — The Piconet

- A piconet is an independent, ad hoc network of at least two devices (DEVs) in radio proximity.
- One device becomes the Piconet Coordinator (PNC), providing synchronization, QoS, power management, and access control.
- The PNC allocates channel time; once allocated, DEVs can talk directly to each other, peer-to-peer.
- Any DEV can become the PNC — the role is negotiated, not fixed hardware.



Solid = PNC control links Dashed = direct peer-to-peer data link

802.15.3 Superframe Structure

- Channel time in a piconet is organized into repeating superframes controlled by the PNC.
- Beacon: carries timing and control information used to synchronize all DEVs.
- CAP (Contention Access Period): command and asynchronous data, accessed via CSMA/CA.
- CTAP (Channel Time Allocation Period): time-division slots (MCTAs / CTAs) guaranteed to specific DEVs — supports QoS.



802.15.3 Physical Layer

Spectrum & channels

- Operates in the unlicensed 2.4 GHz ISM band (2400–2483.5 MHz).
- Defines 5 channels, each 15 MHz wide; a piconet uses one channel at a time.
- Single physical layer shared by all devices in the piconet.

Modulation & data rates

QPSK	11 Mbps
DQPSK	22 Mbps
16-QAM	33 Mbps
32-QAM	44 Mbps
64-QAM	55 Mbps

Higher-order modulation trades range for throughput — 64-QAM delivers the fastest 55 Mbps link but needs the strongest signal.

802.15.3 MAC Layer Features

1 Quality of Service

Guaranteed Time Slots (GTS) in the CTAP reserve bandwidth for streaming audio/video.

2 Security

Optional AES-128 based encryption and authentication protects piconet data.

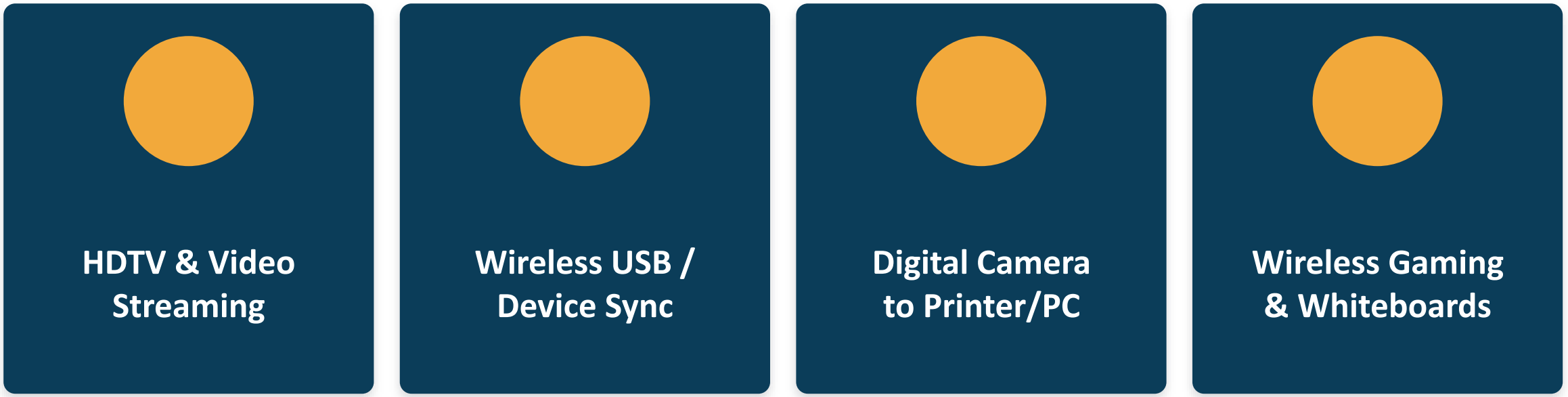
3 Power Management

DEVs use sleep and active modes negotiated with the PNC to save energy.

4 Ad Hoc Formation

Piconets form and dissolve dynamically with no fixed infrastructure.

802.15.3 Applications



Common thread: all require sustained high throughput over a short range — exactly what 802.15.3's TDMA-based CTAP guarantees.

IEEE 802.15.4 — Overview

20–250	10–100 m	2003	Years
kbps data rate	operating range	year ratified	battery lifetime

Design philosophy

- Optimized for low cost, low power, and low data rate — the opposite trade-off from 802.15.3.
- The MAC and PHY foundation for ZigBee, 6LoWPAN, Thread, and WirelessHART.
- Devices should run for months or years on small batteries, so the protocol favors idle/sleep time.

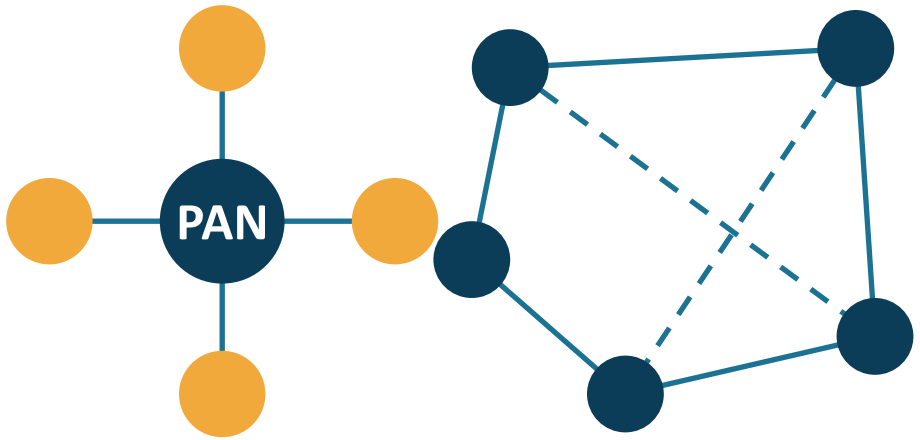
Device Types & Network Topologies

Device types

- FFD (Full Function Device) — can act as coordinator, router, or end device; full protocol.
- RFD (Reduced Function Device) — simple end device, e.g. a sensor; talks to one FFD only.

Topologies

- Star — all RFDs communicate through one central PAN coordinator.
- Peer-to-peer / Mesh — any FFD can talk to any FFD in range, enabling multi-hop routing.
- Cluster-tree — a hybrid, hierarchical combination of star clusters linked in a tree.



A PAN Coordinator (an FFD) is mandatory in every 802.15.4 network — it starts the network and assigns its PAN ID.

802.15.4 Physical Layer — Frequency Bands

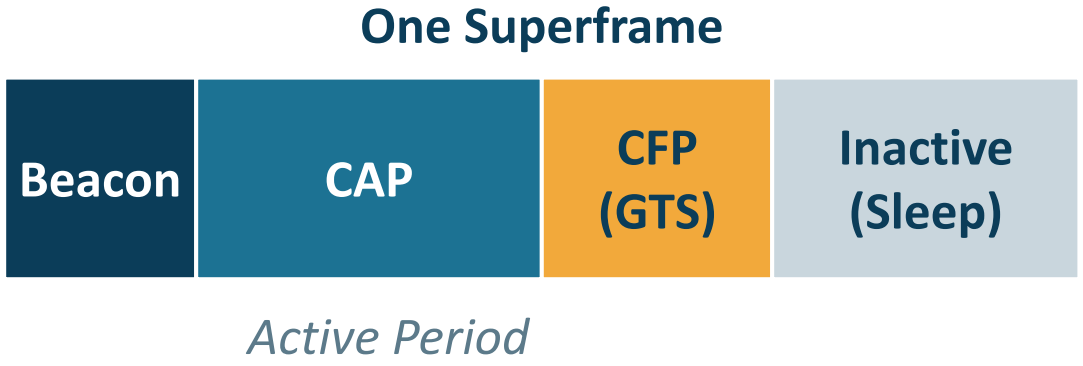
Band	Region	Channels	Rate	Modulation
868 MHz	Europe	1	20 kbps	BPSK
915 MHz	Americas	10	40 kbps	BPSK
2.4 GHz	Global (ISM)	16	250 kbps	O-QPSK

Key takeaway

- The 2.4 GHz band offers the most channels and the highest rate, so it is used worldwide (this is what ZigBee runs on).
- 868/915 MHz sub-GHz bands trade data rate for longer range and better wall penetration.

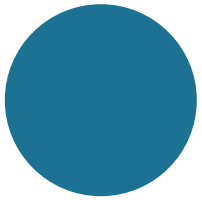
802.15.4 MAC Layer — Channel Access

- Beacon-enabled mode: coordinator sends periodic beacons; devices synchronize to an active/inactive superframe.
- Non-beacon mode: devices use unslotted CSMA/CA and transmit whenever the channel is free — simpler, but less deterministic.
- CFP (Contention-Free Period) offers Guaranteed Time Slots (GTS) for latency-sensitive devices.
- The Inactive Period lets the coordinator and devices sleep, which is the main source of multi-year battery life.

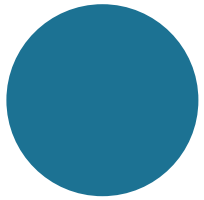


Superframe duration is set by two parameters: Beacon Order (BO) and Superframe Order (SO), both configured by the coordinator.

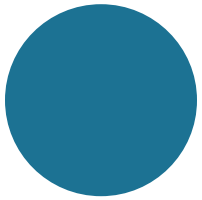
802.15.4 Applications



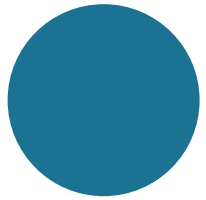
Home Automation (ZigBee)



Industrial Sensing (WirelessHART)



Smart Metering / Smart Grid



Healthcare & Wearable Monitors

Common thread: sparse, small data payloads sent infrequently by battery-powered devices — the exact profile 802.15.4 is optimized for.

COMPARATIVE ANALYSIS

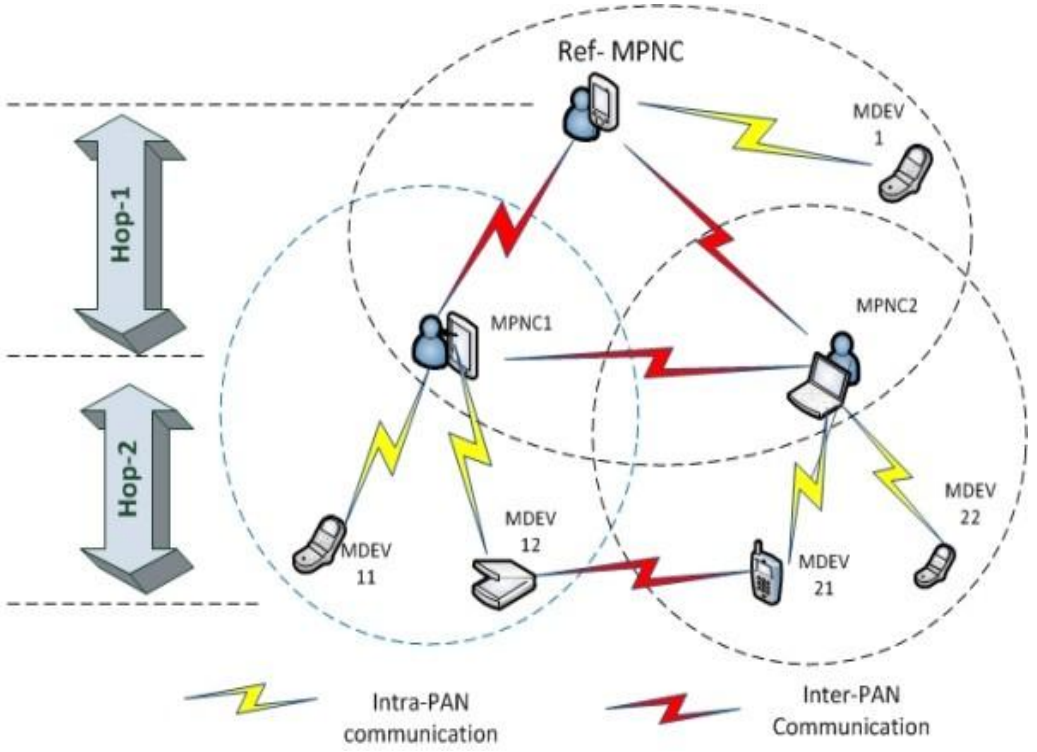
802.15.3 vs 802.15.4 at a Glance

Parameter	IEEE 802.15.3	IEEE 802.15.4
Data rate	11 – 55 Mbps	20 – 250 kbps
Range	~10 m	10 – 100 m
Power use	Moderate	Ultra-low
Topology	Piconet (star + peer)	Star / Mesh / Cluster-tree
Target use	Multimedia streaming	Sensing & control (IoT)
Coordinator	PNC	PAN Coordinator (FFD)

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.5: Mesh Networking**



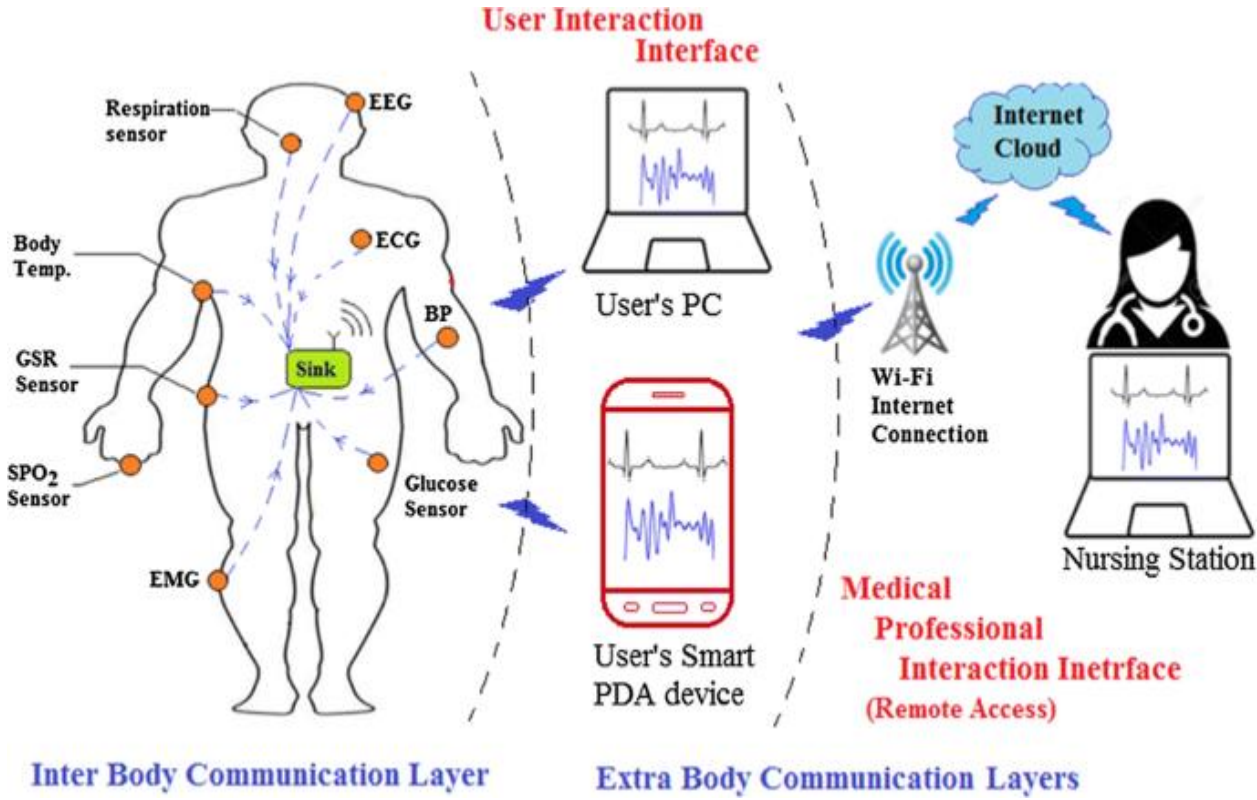
- IEEE 802.15.5 is wireless mesh topologies in local and metropolitan area networks.
- It promotes stable, scalable, and interoperable wireless mesh topologies.
- This standard is composed of two parts:
  - low-rate WPAN mesh and
  - high-rate WPAN mesh networks.
- The common features of both meshes include network initialization, addressing, and multi-hop unicasting.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.6: Body Area Networks**

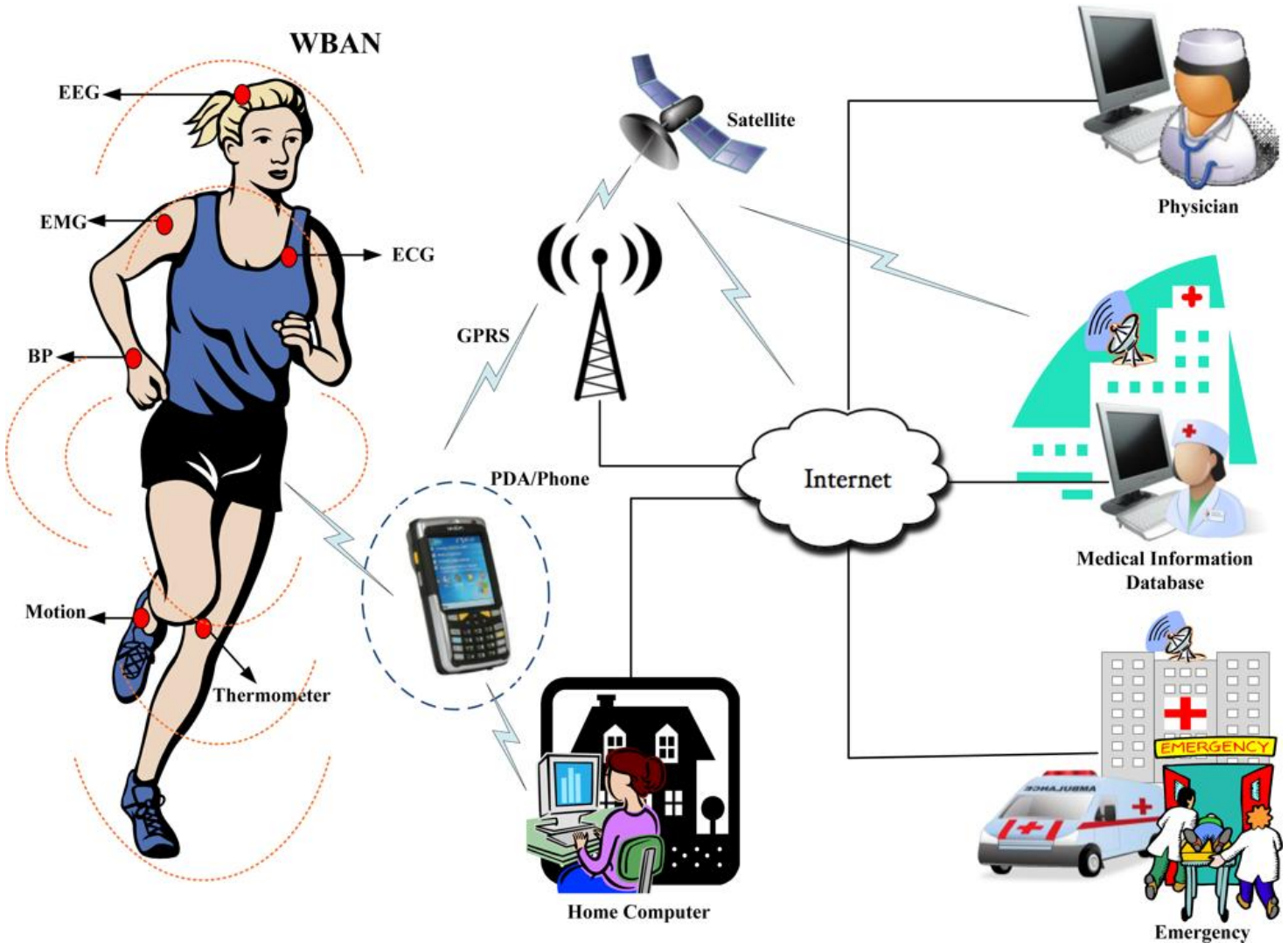
  - low-power and short-range wireless standard to be optimized for devices and operation on, in, or around the human body (but not limited to humans)
  - to serve a variety of applications including medical, consumer electronics, and personal entertainment.



IOT Data link layer and Network layer protocols

IEEE 802.15

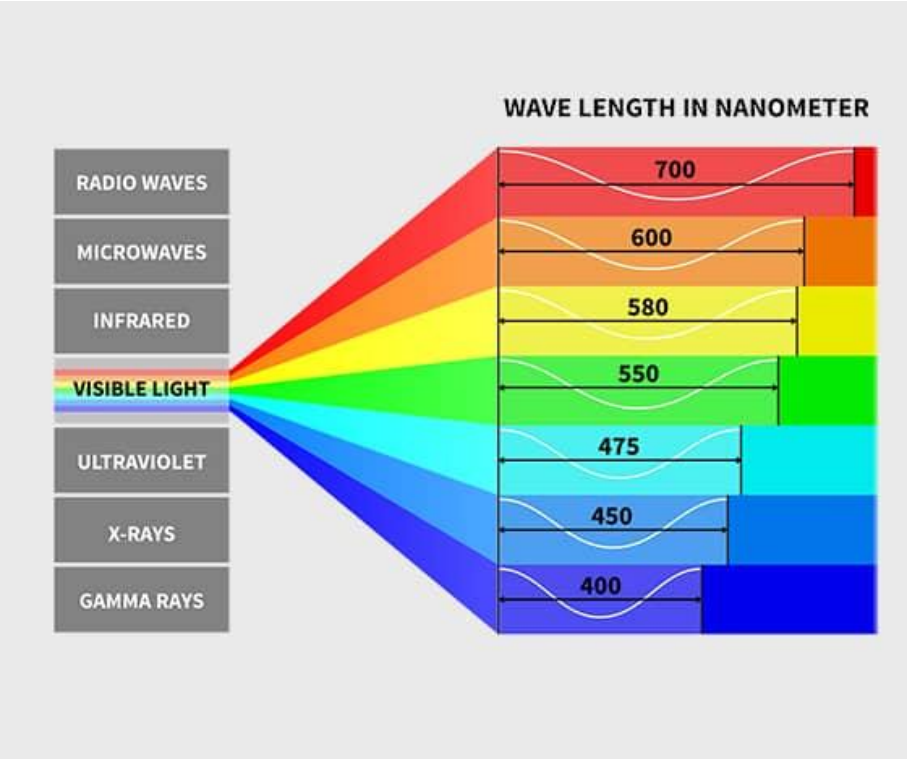
- **IEEE 802.15.6: Body Area Networks**

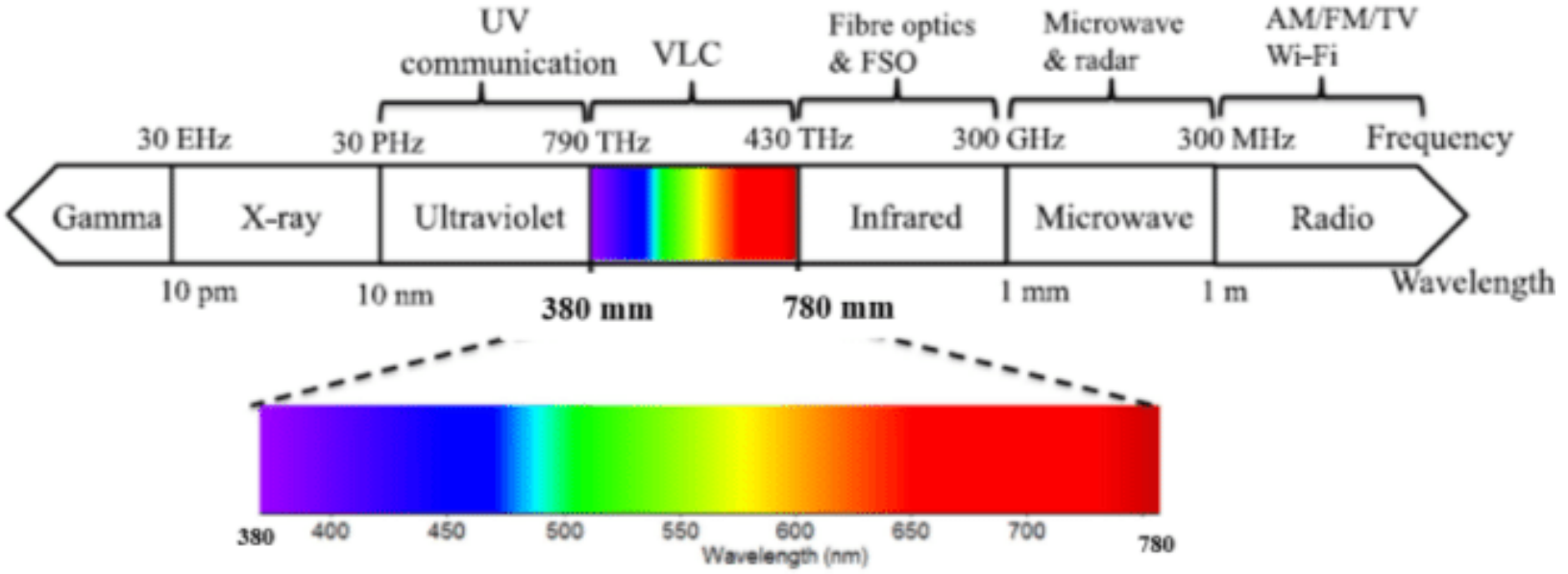


IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.7: Visible Light Communication**





IOT Data link layer and Network layer protocols

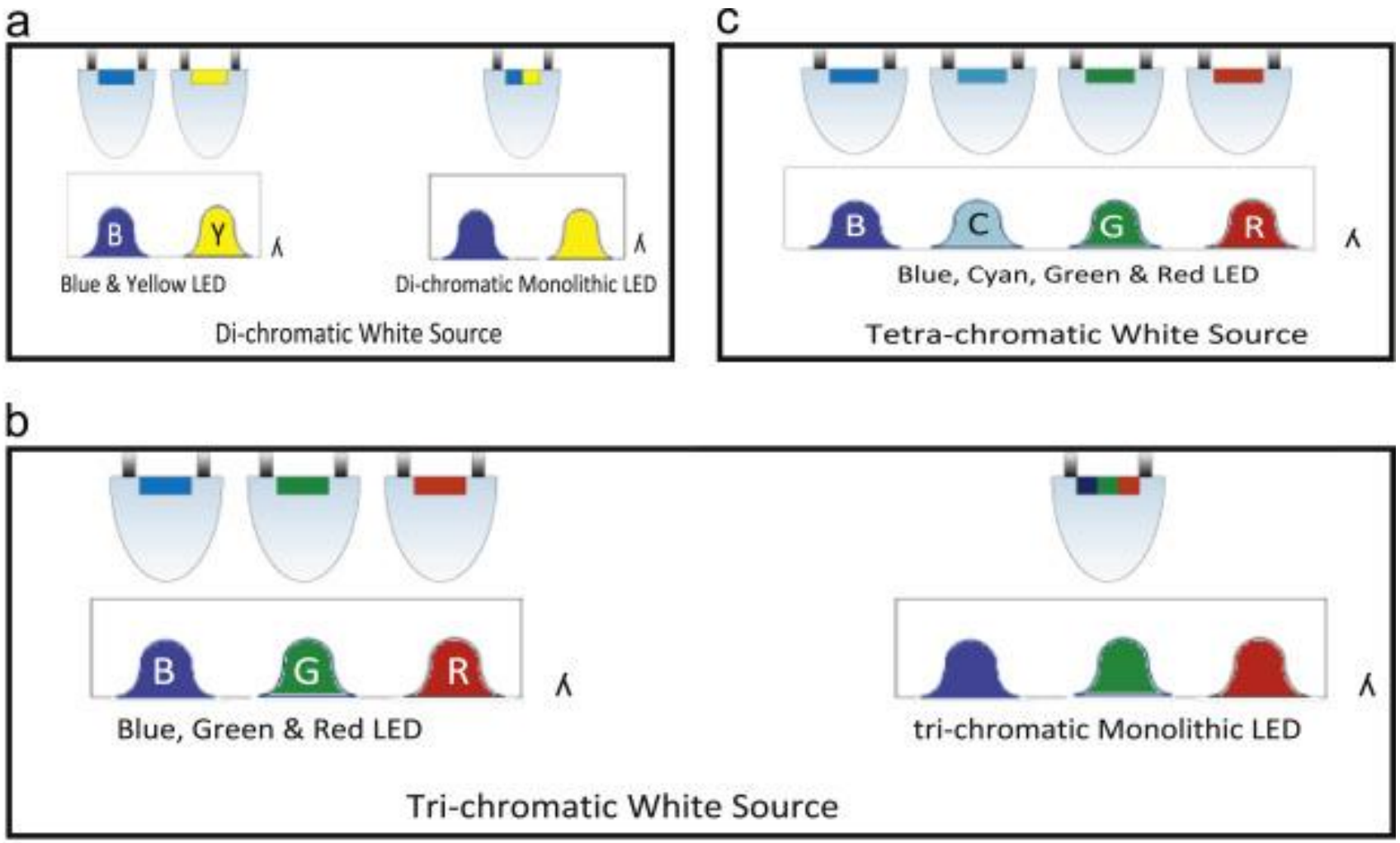
IEEE 802.15

- **IEEE 802.15.7: Visible Light Communication**
- free-space optical communication using visible light (wavelengths between 380 nm and 780 nm).
- to support optical camera communications (OCC) and light fidelity (LiFi).
  - Indoor Communication
  - Outdoor Applications
  - Underwater Communication
  - Healthcare and Aviation

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.7: Visible Light Communication**



IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE P802.15.8: Peer Aware Communications**
- IEEE P802.15.8 received IEEE Standards Board approval on 29 March 2012 to form a Task Group to develop a standard for Peer Aware Communications (PAC) optimized for peer-to-peer and infrastructure-less communications with fully distributed coordination operating in bands below 11 GHz.
- The proposed standard is targeting data rates greater than 100 kbit/s with scalable data rates up to 10 Mbit/s.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE P802.15.8: Peer Aware Communications**

Key Features of IEEE P802.15.8:

1.Network Topology:

1. Supports **peer-to-peer**, **group**, and **mesh networking**.
2. Does not depend on fixed infrastructure, enabling ad-hoc connectivity.

2.Primary Applications:

1. Personal and consumer device communication (e.g., smartphones, laptops).
2. Industrial and military environments for quick and secure device-to-device communication.
3. Emergency response systems, where infrastructure may be unavailable or damaged.

3.Performance Goals:

1. **Low latency**: Critical for real-time applications.
2. **High data rates**: Capable of supporting multimedia applications.
3. **Energy efficiency**: Optimized for devices with limited power resources, such as wearables or IoT devices.
4. **Robustness**: Operates reliably in challenging environments with interference or obstruction.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE P802.15.9: Key Management Protocol**
- IEEE P802.15.9 received IEEE Standards Board approval on 7 December 2011 to form a Task Group to develop a recommended practice for the transport of Key Management Protocol (KMP) datagrams.
- The recommended practice will define a message framework based on Information Elements as a transport method for key management protocol (KMP) datagrams and guidelines for the use of some existing KMPs with IEEE Std 802.15.4.
- The recommended practice will not create a new KMP

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE P802.15.9: Key Management Protocol**

Key Features of IEEE P802.15.9:

1.Purpose:

1. Introduces mechanisms for secure key establishment, distribution, and management for WPANs.
2. Facilitates authentication, encryption, and data integrity in WPANs.

2.Scope:

1. Provides a general framework for incorporating KMPs in WPAN technologies such as IEEE 802.15.4 (used in Zigbee and Thread) and other related standards.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE P802.15.9: Key Management Protocol**

3. Supported Key Management Protocols:

1. IEEE P802.15.9 defines the interface between WPAN devices and KMPs, enabling the integration of commonly used protocols such as:
   1. **Pre-Shared Keys (PSKs)**: Shared in advance between devices.
   2. **Public Key Infrastructure (PKI)**: Uses asymmetric cryptography for secure communication.
   3. **Elliptic Curve Cryptography (ECC)**: Lightweight and efficient cryptographic mechanism.
   4. Other advanced mechanisms that align with the resource constraints of WPAN devices.

4. Security Features:

1. Supports authentication of devices before key exchange.
2. Ensures confidentiality and integrity of the key exchange process.
3. Prevents attacks such as eavesdropping, man-in-the-middle, or replay attacks.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE P802.15.10: Layer 2 Routing**
- EEE P802.15.10 received IEEE Standards Board approval on 23 August 2013 to form a Task Group to develop a recommended practice for routing packets in dynamically changing 802.15.4 wireless networks (changes on the order of a minute time frame), with minimal impact to route handling.
- The goal is to extend the coverage area as the number of nodes increase.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE P802.15.10: Layer 2 Routing**

- The route related capabilities that the recommended practice will provide include the following:
- Route establishment
- Dynamic route reconfiguration
- Discovery and addition of new nodes
- Breaking of established routes
- Loss and recurrence of routes
- Real time gathering of link status
- Allowing for single hop appearance at the networking layer (not breaking standard L3 mechanisms)
- Support for broadcast
- Support for multicast
- Effective frame forwarding

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE P802.15.10: Layer 2 Routing**

Key Features of IEEE P802.15.10:

1.Purpose:

1. Provides routing services for IEEE 802.15 networks at the **data link layer (Layer 2)**.
2. Optimizes data delivery in **multi-hop topologies**, where devices may act as both end nodes and intermediaries for forwarding data.

2.Scope:

1. Designed to work seamlessly with WPAN standards like **IEEE 802.15.4**.
2. Addresses limitations of higher-layer routing protocols (e.g., network-layer routing) by directly integrating routing into the MAC layer for better efficiency and lower latency.

3.Routing Mechanisms:

1. Supports **mesh networking**: Devices communicate with each other directly or via intermediate nodes.
2. Provides mechanisms for:
    1. **Path discovery**: Identifying optimal routes for data delivery.
    2. **Path maintenance**: Handling dynamic changes in network topology due to node mobility, failures, or environmental factors.
    3. **Path recovery**: Quickly adapting to broken routes to ensure reliability.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE P802.15.10: Layer 2 Routing**

Key Features of IEEE P802.15.10:

4. Advantages:

1. **Low latency**: By operating at Layer 2, it eliminates the need for higher-layer routing, reducing processing overhead and latency.
2. **Energy efficiency**: Optimized for low-power devices, making it ideal for battery-operated IoT and sensor networks.
3. **Scalability**: Handles large networks with many devices efficiently.
4. **Robustness**: Maintains reliable communication in challenging environments.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE P802.15.10: Layer 2 Routing**

Key Features of IEEE P802.15.10:

5. Applications:

1. **Internet of Things (IoT)**: Ensures reliable and efficient data exchange in smart homes, industrial automation, and smart agriculture.
2. **Healthcare**: Secures communication in medical devices and patient monitoring systems.
3. **Military and Emergency Services**: Provides robust communication in dynamic and infrastructure-less environments.
4. **Smart Cities**: Supports mesh networks for smart lighting, traffic management, and public safety systems.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.13: Multi-Gigabit/s Optical Wireless Communications**

- The first meeting of Task Group 13 was held during March 2017, aiming at a new standard on light fidelity (LiFi), i.e. mobile communications by using the light.
- The aim is to address industrial applications, i.e. ultra-reliable, low-latency connectivity with negligible jitter for next-generation IoT.
- Compared to 802.15.7, the group decided to rewrite the standard entirely, based on existing and new contributions, to meet those targets.
- The group first worked on a low-power pulsed modulation PHY (PM-PHY) using On-Off-Keying (OOK) with frequency-domain equalization (FDE) and also a high-bandwidth PHY (HB-PHY) based on orthogonal frequency-division multiplexing (OFDM) adopted from ITU-T G.9991.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.13: Multi-Gigabit/s Optical Wireless Communications**

- The group also decided to implement mobility by considering access points in the infrastructure and mobile users in the service area as inputs and outputs of a distributed multiple-input multiple-output (D-MIMO) link.
- 802.15.13 supports D-MIMO natively with a minimalistic design, suitable for specialty applications. It is implementable on low-cost FPGAs and off-the-shelf computing hardware.
- The Working Group letter ballot and the IEEE SA Ballot were started in November 2019 and November 2020, respectively.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.13: Multi-Gigabit/s Optical Wireless Communications**

Key Features of IEEE 802.15.13:

1.Purpose:

1. Designed to provide **multi-gigabit data rates** for optical wireless communication systems.
2. Offers a robust and efficient protocol for scenarios requiring high-speed data exchange over short distances.

2.Medium:

1. Uses **optical wireless communication** as the transmission medium, particularly in the visible, infrared, and ultraviolet spectrum.

3.Performance Goals:

1. **Data Rates**: Targets rates exceeding **10 Gbps**.
2. **Latency**: Optimized for ultra-low latency to support real-time applications.
3. **Range**: Typically supports short distances but aims for reliability in line-of-sight and near-line-of-sight scenarios.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.13: Multi-Gigabit/s Optical Wireless Communications**

Key Features of IEEE 802.15.13:

4. Applications:

1. **Augmented Reality (AR) and Virtual Reality (VR)**: High-speed data transmission for immersive experiences.
2. **High-Definition Media Streaming**: Transmitting ultra-high-definition video content wirelessly.
3. **Wireless Backhaul**: Connecting base stations or access points without cables.
4. **Industrial Automation**: Enabling high-speed and secure data exchange in manufacturing and robotics.
5. **Secure Communication**: Providing high-speed, interference-free links in environments sensitive to RF emissions (e.g., hospitals, aircraft).